

Multimodal World Knowledge: Architectures, Reasoning, and Embodied Intelligence

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Multimodal artificial intelligence has historically prioritized static perception and isolated generative tasks over dynamic environmental understanding. The integration of perceptual inputs with action-conditioned simulation marks a decisive transition toward agents capable of internalizing physical laws and planning long-horizon trajectories. Current methodologies struggle to maintain causal consistency and temporal coherence when navigating open-world complexity, often producing hallucinated states or failing under distributional shifts. This survey presents a unified taxonomy of Multimodal World Knowledge, categorizing advancements into generative latent dynamics, embodied vision-language-action policies, cognitive agentic architectures, and unified multimodal representations. Analysis reveals that achieving reliable autonomy requires moving beyond mere pattern matching to incorporate neuro-symbolic constraints that enforce physical validity within learned latent spaces. Furthermore, existing evaluation protocols frequently overlook the necessity of closed-loop feedback, creating a disconnect between benchmark performance and real-world deployability. Future progress depends on developing self-evolving data ecosystems and native omni-modal foundations that align computational efficiency with rigorous causal reasoning for robust physical interaction.

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1 Introduction

The trajectory of artificial intelligence is currently undergoing a fundamental transition from reactive, perception-only systems to anticipatory architectures capable of internalizing environmental dynamics for long-horizon planning [40, 123]. Contemporary frameworks increasingly bridge the gap between high-level semantic reasoning and grounded physical interaction through bidirectional data flows, enabling agents to align abstract goals with concrete motor commands [40, 46]. This evolution necessitates a unified definition of world knowledge that converges generative video dynamics, implicit representation learning, and cognitive mapping into single computational entities [46, 123]. Such integration moves beyond static retrieval or unimodal analysis, establishing a foundation where models simulate counterfactual scenarios and adapt to lifelong learning loops within complex environments.

The core methodologies can generally be categorized into several primary paradigms. 1) Reasoning mechanisms span from linear prompt-based chains to au-

tonomous graph topologies that enable backtracking and asynchronous staging for test-time scaling [167]. 2) A fundamental taxonomic split separates frameworks modeling physical state transitions from those representing introspective mental states via predictive coding theories within partially observable Markov decision processes [104]. 3) Evolutionary roadmaps further classify architectures by their progression from modular perception units to native omni-modal systems that unify generation and agentic planning through adaptive reinforcement learning [96].

A fundamental tension persists between the drive for high-fidelity dimensional synthesis and the computational latency inherent in optimization-based generative frameworks, where feedforward efficiency often conflicts with the memory bottlenecks required for geometric fidelity [66]. Current architectures frequently conflate superficial pattern matching with genuine causal understanding, resulting in systems that predict visual futures without internalizing physical laws or recovering counterfactual consistency [40]. This adversarial dynamic intensifies as model scale

increases, amplifying hallucination risks in out-of-distribution scenarios rather than guaranteeing robust environmental coherence or preventing physics violations [40, 66]. Consequently, the field faces a continuous technological arms race where improvements in simulation diversity often degrade temporal consistency, leaving evaluative metrics insufficient to bridge the widening gap between synthetic generation and reality.

A persistent fragmentation in current research treats explicit simulation and implicit representation as mutually exclusive design choices, preventing the synthesis of a unified framework that leverages both anticipatory rollouts and scalable predictive coding [123]. Joint architectures frequently struggle to maintain temporal synchronization between high-level semantic planners and low-latency physical simulators, creating fragmented feedback loops that inhibit true lifelong learning [46]. This conceptual ambiguity obscures the specific conditions under which internal world models transition from passive observers to active agents capable of robust physical and social reasoning [46, 123]. Therefore, the primary motivation of this survey is to synthesize these fragmented advancements into a cohesive taxonomy that guides the development of Artificial General Intelligence.

Contributions. In this survey, we make three key contributions: **1) Unified Conceptual Framework:** We establish a single computational architecture that integrates perceptual understanding, generative simulation, and agentic reasoning to form a cohesive basis for multimodal world knowledge, as detailed in Sec. 3. **2) Cohesive Taxonomy:** We propose a structured categorization that divides existing methodologies into four distinct paradigms based on their specific mechanisms for internalizing environmental dynamics, providing a clear map of the field in Sec. 3. **3) Strategic Synthesis:** This work consolidates fragmented advancements to guide the development of Artificial General Intelligence capable of robust physical and social reasoning through dynamic world modeling, with future directions outlined in Sec. 4.

Scope. To ensure a focused analysis of active environmental understanding, this survey bounds its coverage to the methodological intersection of generative world models, vision-language-action policies, and cognitive agentic systems as detailed in Sec. 3. We explicitly exclude purely unimodal approaches and static retrieval methods that lack dynamic environmental simulation capabilities, thereby distinguishing our review from broader perception-only studies. Furthermore, while adjacent fields such as standard model-based reinforcement learning provide

essential context in Sec. 2, they remain outside our primary technical scope to maintain a strict emphasis on the transition from passive perception to active world modeling.

Survey Pipeline. The narrative commences by establishing foundational definitions, historical context, and evaluation protocols in Sec. 2 to ground the theoretical basis of multimodal world modeling. Building upon this groundwork, Sec. 3 presents the core technical analysis, categorizing methodologies into four distinct paradigms: generative latent dynamics, embodied vision-language-action, cognitive agentic architectures, and unified multimodal representation. The discussion concludes in Sec. 4 by synthesizing persistent technical barriers and outlining strategic future directions aimed at advancing artificial general intelligence, before final remarks are offered in Sec. 5.

2 Background

This chapter establishes the theoretical and practical foundations for multimodal world knowledge by systematically defining its core components and evolutionary trajectory. We first formalize the mathematical objectives of world modeling, including state transition prediction and counterfactual reasoning within partially observable environments in Sec. 2.1. The narrative then traces the technological progression from early fusion techniques to modern generative simulators in Sec. 2.2, highlighting critical architectural shifts. To quantify advancements, Sec. 2.3 details the landscape of datasets, metrics, and benchmarks used to evaluate physical reasoning and agentic capabilities. Finally, Sec. 2.4 delineates the boundaries between this field and adjacent domains such as model-based reinforcement learning and neuro-symbolic AI, clarifying their distinct contributions and overlaps.

2.1 Problem Definition

POMDP Formalization for Multimodal Inference.

Grounded in the Partially Observable Markov Decision Process (POMDP), multimodal world modeling requires agents to infer latent states from heterogeneous sensory inputs to navigate environments where full observability is impossible [27, 93, 123]. We define the interaction tuple as $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{O}, \Omega, R, \gamma \rangle$, where the agent maintains a belief state b_t over the latent space z_t conditioned on the observation history h_t . The core optimization objective compresses physical dynamics into this compact representation by maximizing the Evidence Lower Bound (ELBO),

which balances reconstruction fidelity against the regularization of the transition function $p(z_{t+1}|z_t, a_t)$:

$$\begin{aligned} \mathcal{L}(\theta, \phi) = & \mathbb{E}_{q_\phi(z|h)}[\ln p_\theta(o|z)] \\ & - \mathcal{D}_{KL}(q_\phi(z_{t+1}|z_t, a_t) || p_\theta(z_{t+1}|z_t, a_t)). \end{aligned} \quad (1)$$

Here, the reconstruction term $\mathcal{L}_{rec}$ ensures the latent code retains sufficient information to recover observations, while the KL-divergence term $\mathcal{D}_{KL}$ enforces consistency with the learned transition prior under finite-resource constraints [27, 93]. This mathematical substrate naturally bifurcates operational strategies: explicit rollout-based paradigms utilize the transition function in Eq. (1) for counterfactual planning and anticipatory control, whereas implicit representation-based approaches leverage the encoded predictive structure for reactive consistency inference [93, 123]. Such a distinction organizes the subsequent methodological taxonomy, separating systems designed for general-purpose simulation from those optimized for decision-coupled spatial encoding.

Latent Dynamics and Structured Inference. Reframing world modeling as the inference of latent state transitions rather than direct observation prediction resolves the objective mismatch inherent in autoregressive sequence models, which often fail to maintain causal consistency over long horizons [5]. This formulation rigorously distinguishes the observation space $\mathcal{O}$ from the latent state space $\mathcal{S}$, prioritizing persistent state tracking over next-token likelihood maximization through cyclic curriculum optimization. Within this framework, the transition function $\mathcal{T} : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$ maps state-action pairs to successor states, where the choice between natural language and structured representations dictates the trade-off between simulation fidelity and verifiability across physical or digital grounding domains [95]. Addressing partial observability necessitates inferring a compact latent representation that captures unobserved environmental facts solely from incomplete observation histories, thereby enabling the pruning of invalid actions $\mathcal{A}_{valid} \subset \mathcal{A}$ without relying on explicit graph reconstruction mechanisms. Consequently, the mathematical objective shifts from simple generative decoding to optimizing structured inference metrics that enforce consistency constraints on the inferred world state, ensuring the latent dynamics remain aligned with the underlying physical or logical rules of the environment.

Counterfactual and Action-Conditioned Constraints. Formalizing counterfactual reasoning requires treating hypothetical state trajectories τ_{cf} not merely as likelihood maximization targets, but as solutions to an optimization problem constrained by

physical consistency laws. This normative framework specifies that a valid generative simulator must infer latent states such that the resulting trajectory satisfies a physical adherence constraint $\mathcal{C}_{phys}(\tau)$, effectively filtering out dynamically impossible futures [150, 207]. We define this objective formally as:

$$\begin{aligned} \tau_{cf}^* = & \arg \max_{\tau} \mathbb{E}[\log p(\tau|z)] \\ \text{s.t. } & \mathcal{C}_{phys}(\tau) \leq \epsilon, \end{aligned} \quad (2)$$

where z represents the inferred latent context and ϵ denotes the tolerance for geometric inconsistency penalties. Complementing this, action-conditioned generation operates as a closed-loop feedback mechanism where the transition probability $p(s_{t+1}|s_t, a_t)$ restricts the solution space to physically plausible outcomes, thereby distinguishing true world modeling from passive text-to-video synthesis [150, 207]. Within this optimization landscape, visual consistency metrics function as penalty terms that enforce alignment with expected action-conditioned dynamics, ensuring generated sequences remain grounded in the underlying mechanics of the environment without reliance on external engineering interfaces.

2.2 Technological History and Roadmap

Disjoint Encoders and Shallow Fusion. Initial approaches to multimodal processing treated visual and textual streams as independent signals, utilizing separate convolutional backbones and recurrent networks that operated in isolation prior to feature concatenation. These architectures relied on late fusion strategies where modality-specific representations, often pretrained on disjoint objectives, were merged only at the final decision layer through simple arithmetic operations such as element-wise summation or Hadamard products. Subsequent refinements introduced shallow co-attention modules and bilinear pooling techniques to bridge these isolated embeddings, yet such mechanisms remained confined to the classifier interface rather than enabling deep cross-modal interaction within the encoder itself.

Unified Architectures and Native Multimodality. Moving beyond modular late-fusion assemblies, contemporary frameworks employ unified backbones that process heterogeneous signals within a single latent space, thereby removing the informational bottlenecks inherent in frozen encoders [8, 165]. This architectural nativity necessitates specific training regimes, ranging from differential learning rates for progressive unfreezing to unified next-token prediction objectives stabilized by normalization techniques such as QK-Norm and Z-loss. Modern systems further extend

this unification to spatial intelligence by transferring semantic priors from 2D foundation models to 3D geometric backbones through cross-modal knowledge distillation and LiDAR-centric pre-training [165]. Achieving true input-output symmetry enables fluid Multi-to-Multi modeling, allowing these architectures to perform end-to-end planning and generative simulation rather than static perception [8, 165].

From Static Perception to Dynamic Simulation.

Contemporary frameworks increasingly recast language understanding as a hierarchy demanding active environmental interaction and physical grounding, moving beyond static distributional matching to address causal reasoning gaps and social embodiment criteria [14]. To circumvent the computational burdens of high-fidelity 3D rendering, recent approaches employ abstract text environments that retain rich action spaces and formal semantics, utilizing PDDL and STRIPS integration to manage object tree state transitions and compositional inference within scalable agent training loops [78]. Bridging these abstraction layers, modern architectures couple efficient sequence models with external rule systems, enabling counterfactual reasoning and dynamic state prediction that transcends pure statistical autoregression through dual pathway planning and physical logic injection [114].

2.3 Datasets, Metrics, and Benchmarks

Foundational Textual and Interactive World Modeling Benchmarks. Early evaluation protocols transitioned from unstructured interaction logs to datasets that ground partial observations in explicit knowledge graph states, enabling precise measurement of state reconstruction fidelity through metrics such as F_1 score and valid action exact match [7]. Subsequent frameworks integrated external commonsense sources to test an agent’s ability to prune implausible actions, utilizing dynamic subgraph retrieval via Graph Attention Networks to improve sample efficiency and normalized scores [120]. More recent multimodal extensions couple visual-spatial hints with textual commands to benchmark joint reasoning in partially observable navigation tasks, where models fuse CNN-BiLSTM representations to infer room topology and maximize success rates [19].

Multimodal Scientific Reasoning and Causal Progression. Shifting from static question answering to interactive simulation environments has introduced procedural verification mechanisms where action sequences function as executable explanations for scientific concepts. Within a partially observable Markov decision process framework, recent benchmarks uti-

lize parametric task variations covering thermodynamics and circuit logic to assess agents via normalized scores and valid action detection [162]. To enforce logical rigor in academic contexts, advanced multimodal reasoning tasks now employ algorithmic generation of chain-of-thought rationales that verify cross-subfigure relational inferences through Monte Carlo Tree Search and backward-chaining verification [112]. Evaluation protocols have further expanded to quantify spatiotemporal coherence and physical law adherence in sequential image generation, utilizing metrics such as the Envision-Score to address the understanding-generation paradox [152]. Comprehensive video-based assessments integrate multi-modal retrieval agents to benchmark long-chain reasoning across diverse disciplines, isolating auditory and visual perception contributions through counterfactual thinking and future prediction tasks [63, 219].

Structured Knowledge Graphs for Domain-Specific Multimodal Retrieval.

The resolution of cross-modal ambiguity has driven a transition from unimodal knowledge bases to heterogeneous graphs that explicitly link visual percepts with structured semantic concepts. Early approaches relied on simple keyword matching, whereas modern frameworks employ neighbor-aware filtering and hierarchical event annotation to disambiguate concepts within specialized domains such as biomedical imaging [52, 166]. Contemporary construction pipelines further integrate large language models for zero-shot concept extraction alongside hybrid rule-based filtering, ensuring high-fidelity triplet grounding across diverse modalities including audio and video through mechanisms like retrieval checkers and k-core decomposition [124, 126]. This structural rigor extends to instruction synthesis, where hierarchical domain ontologies facilitate the generation of scientifically accurate training data that bridges sparse field observations with unified semantic schemas, as demonstrated by consensus-driven validation in oceanographic contexts [190]. Evaluation of these systems typically relies on metrics such as *Recall@10*, *MRR*, and *Accuracy* to quantify the fidelity of sonar-optical alignment and inter-annotator agreement.

Embodied Navigation and Egocentric Trajectories. Moving beyond third-person logs, synchronized egocentric datasets now integrate gaze vectors as a primary predictive signal for locomotor intent prior to kinematic execution [201]. Contemporary evaluation protocols enforce strict vision-only partial observability by decoupling agent policies from specific physics engines, verifying task success through terminal-state scripts rather than static path matching across heterogeneous 3D backends [48]. Furthermore, assessment

frameworks for household agents decompose object selection into sequential filtering stages that evaluate utility, contextual norms, and abstract physical state variables under non-ideal conditions [2].

Advanced Metrics for Multimodality, Robustness, and Atomic Capability Decomposition. Moving beyond simple likelihood scores, early approaches to stochastic dynamics adopted entropy-based measures to explicitly quantify the separation and weight distribution of mixture components in latent transition models. Techniques such as Mixing Coefficient Entropy (MCE) and Weighted Average KL Divergence (WAKLD), alongside Jensen-Shannon Divergence and Self Earth Mover’s Distance, provided granular insights into Gaussian mixture outputs under action-masking protocols, particularly revealing sensitivity when component counts remained below six [136]. Contemporary frameworks have since shifted toward decomposing world modeling into orthogonal perceptual and predictive dimensions, utilizing controlled counterfactuals to isolate specific reasoning failures rather than relying on aggregate task success. Systems like WM-ABench employ a Task-to-Quiz formulation grounded in weighted set cover theory to generate trajectory-conditioned ground truth, while Relative Entanglement metrics and Environment Understanding Scores measure the divergence between Task Success Rates and filter non-answerable questions to pinpoint capability gaps [49, 107]. Rigorous robustness assessment now distinguishes between absolute performance degradation and relative stability by incorporating real-world capture corruptions and subtle typographic attacks alongside synthetic noise. Benchmarks such as R-Bench and the SCAM dataset calculate Absolute Robustness (R_a) and Relative Robustness (R_r) using reference-distorted pairs, evaluating model resilience against in-the-wild corruptions and Subtle Character Attacks through zero-shot cosine similarity and prediction shift measurements [89, 175]. Finally, comprehensive benchmarking enforces cross-modal complementarity through ablation-based screening, replacing heuristic scoring with deterministic checklist protocols to verify atomic capability adherence. Protocols like FysicsWorld utilize Cross-Modal Complementarity Screening and AEGIS to enforce any-to-any input-output consistency, while Deterministic Checklist-based Evaluation applies binary judgments to TTS-ASR semantic alignment, validated through cross-task correlation analysis and human agreement metrics [81, 101].

2.4 Related Research Domains

Grounded Representation and Embodiment. Theoretical frameworks posit that semantic understanding cannot arise solely from distributional statistics but requires ascending a hierarchy of environmental interaction scopes [14]. This progression necessitates shifting from passive text observation to active perception, physical intervention, and social negotiation capabilities to construct true meaning. Current large-scale language models remain confined to lower grounding levels within this taxonomy, fundamentally limiting their generalization to novel physical scenarios and social contexts due to absent social identity representation and simulation fidelity barriers.

Architectural Convergence in Latent Simulation. Moving beyond raw pixel prediction, modern model-based reinforcement learning increasingly adopts unified amodal latent spaces that facilitate cross-modal translation and information broadcasting [116]. Within these shared representations, agents perform mental simulation via imagined trajectories, a strategy that drastically reduces sample complexity compared to standard environment-interaction loops by decoupling policy updates from real-time data collection. Furthermore, dynamic fusion mechanisms operating in this latent domain confer zero-shot robustness to missing sensory modalities, preventing total policy collapse during sensor failure while maintaining performance above critical return thresholds.

Neuro-Symbolic Integration for Probabilistic Physical Reasoning. By decoupling neural perception of multimodal cues from symbolic hypothesis generation, modern approaches resolve physical ambiguity through modular pipeline architectures that facilitate rigorous Bayesian belief updates [204]. These systems integrate probabilistic programming with large language models to enforce constraint satisfaction via rejection sampling over continuous geometric and acoustic likelihoods, effectively marginalizing posteriors across visual fit constraints and CLAP-derived audio signals. Such hybrid inference mechanisms capture non-linear sensory correlations to achieve human-level alignment in occluded object localization tasks, succeeding where pure neural baselines fail to preserve physical consistency.

Structural Knowledge Organization via Graph Construction and Cross-Domain Fusion. Moving beyond static ontologies, recent methodologies employ large language models to drive dynamic graph construction, resolving semantic ambiguity through embedding-based clustering and contextual co-occurrence weighting. By utilizing zero-shot concept extraction alongside dense vector representa-

tions and density-based clustering, these approaches identify cohesive conceptual cores through recursive decomposition and community detection algorithms [124]. To address data scarcity in physical world modeling, cross-domain fusion frameworks establish alignment principles that differentiate between precise explicit links and coarse implicit connections for integrating heterogeneous data sources. These architectures utilize multiview alignment and dependency principles to preserve spatio-temporal integrity, effectively distinguishing between fine-grained causal chains and broader associative patterns during root cause analysis [231].

2.5 Discussion

Despite the proliferation of specialized benchmarks, current evaluation frameworks disproportionately prioritize superficial pixel fidelity and static accuracy over the causal consistency and dynamic stability required for reliable long-horizon planning [93, 98]. This metric misalignment obscures critical failure modes, as standard uncertainty quantification methods frequently conflate distinct sources of variance, failing to isolate true multimodal transition dynamics essential for stochastic environment modeling [116, 136]. Consequently, a fundamental theoretical disconnect persists between high-fidelity spatial representations and efficient temporal reasoning, forcing artificial trade-offs between geometric precision and computational scalability that hinder the development of unified agents [78, 124]. Without standardized definitions that decouple epistemic uncertainty from aleatoric noise or reconcile static network topologies with dynamic state transitions, cross-paradigm comparison remains fragmented and reproducibility elusive.

3 Methods

This section categorizes multimodal world knowledge methodologies into four distinct conceptual paradigms based on their primary mechanism for internalizing environmental dynamics. [Sec. 3.1](#) examines architectures that learn to predict future states and transitions through generative modeling, enabling mental simulation and counterfactual reasoning. [Sec. 3.2](#) covers approaches that integrate perception, reasoning, and control into unified agents capable of interacting with physical environments via vision-language-action policies. High-level reasoning systems utilizing memory, tools, and search strategies for complex multi-step planning are analyzed in [Sec. 3.3](#). Finally, [Sec. 3.4](#) discusses foundational meth-

ods for learning shared embedding spaces that align heterogeneous modalities to support downstream reasoning and generation tasks.

3.1 Generative Latent Dynamics

Generative latent dynamics establish a framework for agents to anticipate future states and environmental transitions through internal simulation, facilitating both mental rehearsal and counterfactual reasoning. This section first examines video-based state prediction methods that forecast visual frames directly from actions or textual conditions in [Sec. 3.1.1](#). We then analyze approaches that compress sensory inputs into compact latent representations to enable efficient trajectory modeling and planning in [Sec. 3.1.2](#). The discussion extends to diffusion-based world simulation techniques designed to generate diverse and physically consistent future scenarios in [Sec. 3.1.3](#), followed by an investigation of autoregressive token prediction architectures that unify world modeling as a next-token task across interleaved modalities in [Sec. 3.1.4](#). Finally, [Sec. 3.1.5](#) explores hybrid generative architectures that integrate discrete and continuous representations to combine symbolic reasoning with neural generation capabilities.

3.1.1 Video-Based State Prediction

Architectural Strategies for Long-Horizon Temporal Consistency. Mitigating autoregressive drift in extended video generation necessitates periodic context refreshment mechanisms that bound error accumulation without requiring model retraining, often implemented via sliding window re-encoding or latent state updates with negligible latency overhead [176]. To prevent representation collapse while avoiding the token cost of explicit textual reasoning, auxiliary feature matching aligns internal latents with future visual dynamics through dense supervision and stop-gradient future features [200]. Scalable reinforcement learning post-training further corrects semantic misalignment in pixel-level objectives by distilling high-capacity multimodal judges into lightweight student reward models optimized via policy gradient methods [176]. Finally, parallel latent refinement rounds constrain pixel-level drift in continuous state spaces, utilizing Jacobi-style iterations to satisfy bounded tail error constraints over extended horizons [200].

Disentanglement of Static Backgrounds and Object-Centric Dynamics. Explicit attention mechanisms isolate static environmental context from dynamic object states by cropping local background patches, thereby enforcing situation-aware behavior without requiring global re-encoding [103]. To

Table 1 Taxonomy of representative methods in **Generative Latent Dynamics**. This table presents a taxonomy of generative world models, shows their inference mechanisms and state representations, and highlights the prevalence of hybrid autoregressive-diffusion strategies.

Method	Venue	Inference Mode	State Space	Highlight
Latent Space Trajectory Modeling (Sec. 3.1.2)				
LatentUM [83]	arXiv’26	Latent Autoregressive	Semantic CLIP	MBAQ preserves semantic fidelity in discretized latent spaces, avoiding pixel reconstruction for cross-modal reasoning.
AffectVerse [223]	arXiv’26	Rollout Prediction	Latent Belief	Cross-modal temporal imagination rolls out latent audiovisual tokens to construct robust belief states from partial observations.
Lumo-2 [147]	arXiv’26	Block Autoregressive	Latent Dynamics	Latent world dynamics space encodes physically grounded futures, replacing explicit planning with implicit geometric reasoning.
Diffusion-Based World Simulation (Sec. 3.1.3)				
COMBO [211]	ICLR’24	Diffusion Search	Video Point Clouds	Factorizes joint actions into text components for composable multi-agent video diffusion without retraining.
GEM [60]	CVPR’25	Conditional Diffusion	Latent Video	Injects sparse semantic tokens and dynamic noise schedules to control disentangled elements in long-horizon synthesis.
UniDriveDreamer [224]	arXiv’26	Joint Diffusion	Aligned Latents	Aligns LiDAR and RGB latent distributions via Unified Latent Anchoring for single-stage cross-modal generation.
Autoregressive Token Prediction (Sec. 3.1.4)				
OmniJARVIS [172]	NeurIPS’24	Autoregressive	Discrete Behavior Vector	FSQ discretizes actions into semantic tokens for unified autoregressive vision-language-action modeling.
DiDA [33]	arXiv’25	Discrete Diffusion	Quantized Textual	Discrete Diffusion Adaptation accelerates visual token generation within unified autoregressive transformer frameworks.
ITWM [94]	ACL’25	Next-Token Prediction	States	Supervised fine-tuning internalizes environment dynamics via multi-turn next-state prediction in ReAct frameworks.
Hybrid Generative Architectures (Sec. 3.1.5)				
Graph World Model (GWM) [45]	ICML’25	Graph Message Passing	Multimodal Graph	Graph nodes integrate embeddings and text, routing queries to LLM or diffusion decoders for unified state transitions.
ThinkMorph [53]	arXiv’25	Interleaved Generation	Text-Image Tokens	Delimiter tokens govern autonomous switching between text reasoning and image generation within interleaved multimodal sequences.
Cosmos 3 [1]	arXiv’26	Dual-Stream Hybrid	Unified Tokens	Dual-tower architecture routes autoregressive reasoning and diffusion generation streams via modality-specific encoders and joint attention.

capture multimodal futures that exceed unimodal Gaussian assumptions, hierarchical latent variable structures decouple state dynamics from attribute generation through versatile propagation modules and object-state recurrent networks. Furthermore, depth-aware graph neural networks integrate spatial relational inductive biases to jointly resolve object-object interactions and occlusion events, utilizing specialized propagation units to model collision constraints effectively [103].

Action-Conditioned Physical Consistency. To bridge static perception with dynamic control, surrogate trajectory prediction tasks enforce physical plausibility by training models to anticipate future point motions directly from unlabeled video data [195]. This approach utilizes coordinate quantization and temporal reasoning to ground visual dynamics without explicit action labels. Extending this foundation, action conditioning ensures kinematic feasibility by aligning generated pixel trajectories with robotic command constraints through direct spatial grounding rather than discrete symbolic mapping [195]. Computational efficiency during such generation is managed through dynamic routing mechanisms that selectively activate video forecasting modules only during fine-grained manipulation phases, thereby optimizing the

trade-off between inference latency and physical fidelity [148]. Further ensuring semantic alignment, these systems interleave textual subtask guidance with visual dynamics modeling to maintain precision in low-level state transitions through adaptive reasoning and flow matching objectives.

Computational Costs of Pixel-Space Optimization. Achieving high-fidelity geometric and dynamic consistency in video-based state prediction often necessitates optimization-based frameworks operating directly in high-resolution pixel spaces, yet this approach incurs prohibitive computational latency compared to feedforward alternatives. Methods employing Score Distillation Sampling (SDS) and iterative refinement loops demonstrate superior physical realism and temporal coherence by explicitly optimizing scene parameters against diffusion priors [66]. However, the requirement for extensive backpropagation through time and the management of substantial memory overhead create significant bottlenecks for real-time application. These trade-offs highlight a fundamental tension where gains in visual fidelity come at the expense of inference speed, limiting the scalability of such techniques in interactive environments.

Visual Fidelity as a Regularizer for Sim-to-Real

Transfer. Explicitly interleaving visual generation steps within reasoning chains mitigates the representational bottlenecks inherent in verbal-only world models for physical and spatial tasks [179]. This architectural choice facilitates flow-matching loss optimization and precise state observation tracking in multi-observable MDPs, significantly improving sample efficiency. However, current unified multimodal architectures exhibit a disconnect where latent causal knowledge fails to fully constrain sequential policy adherence in dynamic processes [152]. Evaluation metrics indicate that while these systems capture causal event progression, they often struggle with spatiotemporal consistency and strict physical law adherence without weighted scoring formulations. Consequently, visual fidelity acts as a necessary regularizer for complex spatial dynamics, particularly when high-fidelity novel view synthesis is required, yet it provides diminishing returns for simple environments requiring only implicit state tracking [152, 179]. Automated judge protocols confirm this visual superiority hypothesis, highlighting a persistent gap between model capabilities and task saturation in low-complexity domains.

3.1.2 Latent Space Trajectory Modeling

Latent Geometry for Counterfactual Arithmetic.

Physical interaction without linguistic supervision induces a latent geometry where vector addition and subtraction directly encode counterfactual state transitions, enabling linear arithmetic on spatial features [44]. Brownian Bridge regularization enforces monotonic relationships between semantic gaps and temporal distances, creating straight-line trajectories in the latent space that support valid interpolation and arithmetic operations independent of rigid goal frames [221]. Constructing an implicit latent world dynamics space bridges high-level reasoning and low-level control by aligning action tokens with physically grounded future possibilities rather than raw signal fidelity [147]. Furthermore, utilizing two-dimensional latent state representations enhances spatial coherence for multimodal sensor fusion, outperforming one-dimensional vectors in predicting both raw sensory data and actionable 3D occupancy [15].

Regularization for Latent Stability. To prevent trajectory collapse during parallel world simulation, orthogonality-based regularization enforces hypothesis separation within continuous latent spaces by optimizing learnable embeddings to maintain distinct representations across multiple diverse hypotheses [69]. Quantifying the resulting multimodal transition distributions requires entropy-driven metrics distinct from unimodal baselines, specifically employing Mixing Coefficient Entropy and Weighted Average KL Di-

vergence to measure component divergence in Gaussian mixture models under action-masking protocols [136]. Preserving local topological structures amidst missing modalities relies on neighbor-based latent alignment, which utilizes availability-aware attention and stop-gradient operations to transfer structural information without inducing representation collapse [158]. Furthermore, stabilizing long-horizon rollouts necessitates quantization objectives that act as a regularizer against latent space contraction, ensuring compressed representations retain sufficient variance to support extended temporal prediction without succumbing to trajectory collapse [83].

Multimodal Fusion and Missing Modality Robustness.

Hybrid objective functions combining contrastive mutual information estimation with selective reconstruction optimize latent fidelity by filtering high-dimensional visual noise while preserving low-dimensional proprioceptive dynamics [12]. Dynamic fusion weight adjustment within a unified amodal Global Workspace enables zero-shot robustness to sensor dropout by re-normalizing contributions from available unimodal VAE latents during mental simulation [116]. Hierarchical attention mechanisms aggregate asymmetric inter-agent observations into a shared latent belief state, allowing decentralized execution to persist despite partial observability and individual sensor failures [4]. Furthermore, modality-aware multi-step attention aggregates cross-modal imagined tokens into dynamic belief queries, conditioning downstream reasoning on predicted affective transitions rather than static sensory inputs [223].

Dimensionality Trade-offs and Counterfactual Reasoning.

The impact of latent dimensionality manifests as a direct trade-off between compression rate and planning horizon accuracy, where overly compact representations sacrifice the fidelity required for stable multi-step lookahead [5]. Discrete linguistic latent spaces enable counterfactual trajectory evaluation without environment interaction by decoupling abstract simulation from concrete execution, effectively mitigating error propagation in long-horizon tasks through hierarchical action separation rather than raw token prediction [37]. To prevent simulation inertia during this reasoning, privileged-future distillation allows models to arbitrate between abstract priors and noisy visual rollouts via advantage-weighted self-distillation and rollout verification gating [237]. This calibration ensures that retry mechanisms activate primarily when abstraction confidence declines, balancing the computational efficiency gained from high compression against the physical fidelity necessary for valid counterfactual analysis.

3.1.3 Diffusion-Based World Simulation

Multimodal Conditioning Architectures. Resolving alignment ambiguities in multi-agent scenarios requires spatially-aware attention mechanisms that explicitly bind textual semantics to specific trajectory regions within the visual latent space [157]. To manage competing structural demands, adaptive fusion architectures employ spatiotemporal weighting maps that dynamically modulate the influence of distinct control branches, enabling region-specific adherence to dense constraints such as edges or depth while preserving generative freedom elsewhere [6]. Further stabilizing these joint processes, unified latent anchoring strategies align heterogeneous sensor distributions into a shared representation through affine transformations, facilitating bidirectional cross-modal consistency without relying on cascaded generation stages [224].

Physical Priors and Constraint Guidance. Embedding partial differential equation solutions as structured conditional priors restricts the diffusion sampling trajectory to physically plausible phase spaces, effectively preventing non-causal dynamic artifacts [233]. This approach integrates PDE-derived simulation outputs directly into the generative process, ensuring adherence to fundamental conservation laws without expanding into specific implementation details of individual physical domains.

Diversity Preservation and Mode Collapse Mitigation. To prevent combinatorial explosion in joint action spaces, recent approaches factorize interactions into independent text-conditioned streams that support arbitrary agent counts without retraining [211]. Complementing this structural decomposition, anchor-based trajectory proposal mechanisms explicitly enforce multimodal hypothesis generation to counteract policy myopia and ensure coverage of distinct navigational modes [118]. Empirical evaluations in closed-loop planning suggest that candidate count saturation occurs at approximately seven candidates, balancing visual foresight diversity with computational tractability. Furthermore, dynamic autoregressive noise scheduling modulates temporal corruption levels during sampling to maintain long-horizon consistency while permitting stochastic variation in object dynamics [60]. This adaptive corruption strategy effectively mitigates temporal discontinuities and sustains diverse scenario synthesis over extended horizons without relying on discrete token interventions.

Sampling Efficiency and Latency Reduction. Addressing the computational burden of iterative denoising, recent frameworks integrate prefix attention

masking with key-value caching to eliminate redundant recomputation during continuous action chunk and subgoal synthesis. This architectural adjustment effectively halves inference latency for tasks involving trajectory prediction and image generation within unified diffusion settings [174]. Distinct from sequential autoregressive decoding, other approaches transform visual prediction into a bidirectional parallel process while strictly maintaining causal constraints for text tokens. By employing optimized continuous sampling strategies and sparse masking mechanisms, these systems achieve significant acceleration in scenario generation without compromising generative fidelity [33].

3.1.4 Autoregressive Token Prediction

Tokenization Granularity and Spatial Fidelity. Resolving modality misalignment in unified architectures often requires discretizing continuous action trajectories into semantic tokens that align with standard language vocabularies, a process facilitated by Finite Scalar Quantization to enable prefix language modeling for imitation learning decoders [172]. Beyond explicit action encoding, open-loop behavior cloning can implicitly induce internal visual state representations where compact visual tokens, interleaved with text via VQ-VAE codes, support spatial planning and geometric reasoning without direct supervision [131]. To optimize the efficiency of such multimodal alignment prior to task-specific fine-tuning, early fusion strategies employ native-resolution encoders like MoonViT-3D that utilize patch packing and aggressive temporal compression ratios, such as a factor of four, to facilitate joint pre-training even under zero-vision supervised conditions [11].

Architectural Mechanisms for Long-Context Sequence Modeling. To overcome the sequential latency inherent in standard autoregressive decoding, Discrete Diffusion Adaptation (DiDA) decouples the causal training objective from bidirectional inference mechanisms, enabling parallel visual token generation while maintaining causal constraints for text [33]. This architectural separation yields substantial efficiency gains, reporting up to a twenty-fold inference acceleration during long-horizon modeling tasks. Complementing this parallelization strategy, Parallel-Agent Reinforcement Learning (PARL) architectures address context exhaustion by transforming task complexity scaling from linear sequential dependencies into concurrent execution patterns [11]. By optimizing orchestrator policies over frozen sub-agent trajectories, these systems isolate critical decision steps and mitigate the accumulation of errors typical in extended sequence generation.

Objective Balancing via Architectural Decoupling. Hard-routed Mixture-of-Transformers architectures isolate conflicting optimization gradients between multimodal understanding and generative synthesis by decoupling parameter subspaces through dual expert routing and modality-specific gating [35, 153]. Complementary interleaved reasoning chains replace isomorphic token prediction by dynamically switching generation modes based on task complexity, utilizing delimiter-controlled transitions to alternate between MSE loss for visual tokens and negative log-likelihood for text [53]. Domain-native decoder heads further preserve geometric validity in scientific outputs while a shared backbone encodes heterogeneous constraints through structured reasoning supervision and task-token routing [225]. Hybrid loss formulations combining flow-matching objectives with autoregressive language modeling resolve scaling asymmetries between data-hungry vision and parameter-efficient text domains, integrating rectified flow with next-token prediction to balance sparsity-induced scaling [35, 153].

Stabilizing Heterogeneous Streams. To counteract catastrophic forgetting during the joint optimization of interleaved modalities, adaptive curriculum sampling dynamically modulates task difficulty ratios based on online performance metrics, thereby preserving gradient utility across domains with disparate signal strengths [64]. This approach utilizes reinforcement learning with curriculum sampling (RLCS) and ratio-based exponential moving averages to align difficulty levels, ensuring that weak subdomain signals do not induce training collapse. Complementing this data-level strategy, token-level self-introspective decoding addresses representation drift by pruning low-importance visual tokens within early decoder layers, which isolates and suppresses vision-text association hallucinations without compromising visual grounding [77]. By employing context-aware token selection and rank-correlation filtering, such mechanisms prevent language-heavy optimization phases from erasing previously learned multimodal alignments. Furthermore, unified reward systems establish a shared interface across diverse tasks, facilitating cross-task knowledge transfer where domain-specific verifiers and anti-hacking constraints ensure that joint optimization yields mutual reinforcement rather than interference [33, 64].

Scaling Dynamics and Data Regimes. Architectural sparsity via Mixture-of-Experts routing resolves fundamental scaling asymmetries where visual modalities demand massive data volumes while linguistic tasks require extensive parameter capacity [153]. This structural regulation enables efficient computation

near 10^{21} FLOPs by decoupling modality-specific processing paths, preventing the saturation observed in dense architectures. Shifting from prompting to supervised fine-tuning on interaction trajectories further advances the data-efficiency frontier, allowing models to internalize environment dynamics rather than relying on brute-force scale for generalization in open-ended domains [94]. Such trajectory-based optimization yields robustness against distribution shifts and expands behavioral coverage beyond the limits of few-shot context windows. Furthermore, pre-training on naturally interleaved video-text sequences at tera-token scales, specifically corpora exceeding 13T tokens, induces spatiotemporal reasoning capabilities that fragmented short-clip or paired-data regimes fail to achieve [33]. This native multimodal interleaving is critical for establishing long-horizon temporal consistency and fostering causal inference without explicit structural priors.

3.1.5 Hybrid Generative Architectures

Architectural Mechanisms for Discrete-Continuous Alignment. To resolve the modality gap, hybrid systems map heterogeneous inputs into a shared geometric latent space where discrete tokens and continuous signals co-exist under a unified temporal coordinate system [1]. This alignment utilizes unified action tokenization and absolute temporal modulation, often employing 3D Multimodal RoPE to synchronize data streams with differing sampling rates. Hierarchical decomposition further stabilizes training by separating high-level abstract reasoning, managed by discrete autoregressive backbones, from low-level perceptual dynamics predicted via continuous diffusion modules [187]. Such stratification prevents representational collapse through vocabulary expansion and sequence lengthening, while a diffusion-based embedding predictor optimizes generative reconstruction losses grounded in observation objectives. Distinct from neural-only approaches, structured probabilistic frameworks achieve alignment through the sequential composition of discrete Hidden Markov Models for logic and continuous switching linear dynamical systems for physical state transitions [31]. These models rely on variational Bayesian inference to handle action-dependent transitions within a four-type structural depth hierarchy, enabling quasi-lossless compression without requiring extensive neural parameterization.

Dynamic Routing and Adaptive Modality Switching. Lightweight gating mechanisms predict binary control tokens derived from pooled multimodal embeddings to conditionally activate specific reasoning

modules only when contextual complexity demands them [148]. Such systems default to action-only decoding unless the router explicitly triggers visual or textual processing branches, thereby reducing computational overhead during simple state transitions. Unified architectures further enforce autonomous mode switching between interleaved and text-only reasoning chains by treating modalities as complementary rather than isomorphic representations [53]. These frameworks utilize delimiter-controlled transitions optimized via dual MSE and NLL objectives to adapt the reasoning chain depth based on task complexity. Extending this compositional logic, Mixture-of-Experts constructs task-relevant contexts through the iterative invocation of atomic primitives, such as distinct “text-think” or “visual rollout” operations [194]. Heterogeneous projections aggregate within a shared workspace before final conditioned decoding, where the output is generated as a function of the input context, allowing the model to synthesize diverse capability sets dynamically without maintaining a monolithic representation space.

Neurosymbolic Integration for Interpretability and Constraint Satisfaction. By decoupling high-level planning from low-level perception, symbolic abstraction layers optimize compact predicate subsets derived from noisy neural proposals to maintain logical consistency across long-horizon tasks [9]. This approach utilizes vision-language model-driven predicate generation followed by hill-climbing sub-selection and PDDL operator learning, effectively filtering noise to enhance planning efficiency. Complementing this structural separation, probabilistic programming frameworks enforce physical constraints by integrating neural sensory priors with Bayesian hypothesis testing, resolving ambiguities in occluded or multimodal scenarios [204]. Such systems employ large language model-generated physical priors and rejection sampling within a Stirling number hypothesis space, leveraging audio likelihoods for posterior marginalization. Furthermore, graph-based state representations unify heterogeneous modalities through embedding-level message passing, allowing action nodes to query structural relationships without expanding token context windows [45]. These architectures utilize multi-modal graph nodes and an action-as-query mechanism, where GCN aggregation and embedding similarity routing satisfy homophilous graph constraints to streamline decision-making.

Optimization Stability in Dual-Objective Regimes. Conflicting gradient signals inherent to hybrid autoregressive-diffusion backbones are resolved by decoupling optimization paths through hard-routed expert subnetworks that physically isolate under-

standing and generation parameters [35]. This architectural separation prevents objective interference while maintaining a unified inference stream, as demonstrated by systems employing Mixture-of-Transformers with distinct dual experts operating over SigLIP2 encodings and FLUX VAE latent spaces. Complementing this structural approach, unified reinforcement learning frameworks stabilize interleaved training by computing shared trajectory-level advantages that propagate credit across heterogeneous modality-native policy gradients [67]. Such methods utilize Group Relative Policy Optimization to align two-level Markov Decision Processes, effectively bridging turn-level macro-actions with step-level micro-actions through dense process rewards without requiring separate value functions for each modality.

3.1.6 Discussion

Architectural decisions in generative latent dynamics ultimately resolve a fundamental tension between physical simulation fidelity and computational latency, where compressing reasoning into latent rollouts or bypassing intermediate trajectory steps yields substantial speedups at the expense of temporal granularity [61, 132]. While diffusion-based approaches secure superior controllability and spatial consistency through modular conditioning and compositional score functions, their reliance on iterative denoising imposes prohibitive overheads for real-time interaction and restricts output resolution [60, 234]. Scaling unified architectures via mixture-of-experts and discrete natural language latents facilitates emergent counterfactual reasoning and long-horizon planning, yet significant barriers persist in interpreting visual layout cues and preventing hallucination during extended sequential rollouts [35, 37, 194].

3.2 Embodied Vision-Language-Action

The integration of perception, reasoning, and control into unified embodied agents requires a cohesive framework that bridges high-level semantic understanding with low-level motor execution. This section examines the architectural components necessary for such systems, beginning with methods that discretize continuous sensory and action spaces into a shared vocabulary for transformer-based training [Sec. 3.2.1](#). We subsequently analyze algorithms that derive policies directly from multimodal trajectories through behavior cloning or reinforcement learning objectives [Sec. 3.2.2](#), alongside approaches that utilize abstract action embeddings to enable hierarchical planning and skill reuse [Sec. 3.2.3](#). The discussion further extends to mechanisms that incorporate real-time

feedback for error correction within the decision loop [Sec. 3.2.4](#) and concludes by evaluating techniques designed to bridge the domain gap between simulated environments and physical deployment [Sec. 3.2.5](#).

3.2.1 Unified Tokenization Strategies

Spectral and Geometric Encoding. Spectral decomposition via Fourier features encodes continuous actions as structured, high-frequency embeddings compatible with autoregressive transformers [\[227\]](#). This approach replaces discrete binning with sinusoidal basis functions and frequency-domain projections, creating a structured embedding space that maintains flow matching compatibility. Distinct from spectral methods, geometric coordinate quantization explicitly bins spatial inputs into discrete tokens. For spatial data, multi-scale geometric conditioning integrates hierarchical 3D point cloud densities into action tokens through bottleneck window attention, preserving local spatial fidelity without disrupting global semantic representations [\[24\]](#). Such mechanisms utilize latent query mechanisms and spatial window partitioning to fuse cross-attention features efficiently.

Hierarchical Quantization and Semantic-Geometric Decoupling. Asymmetric supervision within Residual Vector Quantization isolates coarse motion intent at the primary level while delegating fine-grained geometric residuals to deeper codebook layers, a strategy exemplified by Semantic Residual Quantization (SRQ) which employs asymmetric loss weighting to enforce this intent-residual decoupling [\[86\]](#). Complementing this structural hierarchy, Self-supervised Finite Scalar Quantization replaces continuous latent trajectories with discrete semantic tokens, thereby enabling unified autoregressive modeling alongside language and vision modalities through behavior token vocabulary packing [\[172\]](#). To ensure these discrete representations remain grounded in physical reality, cross-modal contrastive alignment anchors quantized action codes to visual state transitions, effectively disentangling manipulation semantics from background noise prior to policy execution via SigLIP contrastive losses [\[209\]](#).

Unified Token Spaces for Heterogeneous Multimodal Fusion. By treating quantized spatial coordinates as established vocabulary entries, this framework enables joint autoregressive generation across digital and physical tasks [\[195\]](#). Visual perception streams integrate directly into this shared language codebook via specialized boundary tokens, allowing end-to-end transformers to produce mixed-modality sequences without separate decoding heads [\[229\]](#). La-

tent world dynamics further structure this token space by serving as an implicit reasoning bridge that aligns low-level action tokens with high-level semantic representations, ensuring predictive temporal coherence through progressive training objectives [\[147\]](#).

Tokenization Granularity and Inference Latency.

To mitigate the sequential bottlenecks inherent in standard autoregressive generation, block-wise strategies compress multiple generation steps into parallel token blocks while maintaining consistency constraints [\[147\]](#). Distinct from pure discretization approaches, hybrid architectures manage the trade-off between sequence length and trajectory resolution by isolating discrete action chunking through attention masking mechanisms [\[20\]](#). This structural decoupling allows the model to accelerate policy inference via coarse discrete tokens while preserving fine-grained control fidelity through parallel continuous prediction, effectively balancing computational throughput against output smoothness without relying on specific regression loss formulations.

3.2.2 Action-Conditioned Policy Learning

Vision-Agnostic Priors and Expert Distillation.

Decoupling visual grounding from action learning via vision-agnostic pretraining compels policies to internalize robust language-conditioned action priors rather than exploiting superficial visual shortcuts [\[102\]](#). By decomposing trajectories into atomic units and mapping proprioceptive states directly to language instructions, this approach yields a 17.8 percent gain in success rates on downstream tasks. To address parameter competition during multimodal finetuning, dynamic mixture-of-experts architectures employ adaptive routing that isolates reasoning modules from frozen vision-language model weights, achieving a 3.52 times improvement in open-world manipulation [\[238\]](#). Furthermore, interactive imitation learning frameworks distill retrospective feedback from parallel text-world experts to align visual agents with environment dynamics, utilizing algorithms like DAgger-DPO to maintain robustness against noisy observations while minimizing distribution shift [\[198\]](#).

Robustness to Visual Distractions. Policy resilience against sensory noise is achieved through aggressive data augmentation strategies that inject visual perturbations directly into the training stream rather than relying on latent disentanglement [\[155\]](#). By applying background randomization and dynamic occlusion during behavior cloning, these methods force the policy to depend on task-relevant proprioceptive and object-centric features, enabling zero-shot transfer without explicit domain alignment. Dy-

Table 2 Taxonomy of representative methods in **Embodied Vision-Language-Action**. This table presents a taxonomy of embodied vision-language-action methods, shows distinct learning paradigms and world modeling strategies, and highlights architectural diversity in physical reasoning.

Method	Venue	Learning Paradigm	World Modeling	Highlight
Unified Tokenization Strategies (Sec. 3.2.1)				
OmniJARVIS [172]	NeurIPS’24	Behavior Tokenization	Semantic Action Tokens	Applies Finite Scalar Quantization to discretize actions into semantic tokens for unified autoregressive modeling.
Magma [195]	CVPR’25	Joint Pretraining	Trace-of-Mark Forecasting	Maps 2D coordinates and 3D actions to discrete textual tokens via Set-of-Mark coordinate quantization.
WorldBagel [227]	arXiv’26	Supervised Fine-Tuning	Flow Matching Prediction	Employs Fourier Feature Action Tokenizer to structure continuous control inputs within a shared embedding space.
Action-Conditioned Policy Learning (Sec. 3.2.2)				
EMMA [198]	CVPR’24	DAGger-DPO	Parallel Text World	Aligns visual policies via DAGger-DPO using retrospective LLM expert feedback from parallel text worlds.
OpenWebVoyager [62]	ACL’24	Iterative Self-Improvement	Accessibility Tree POMDP	Refines web navigation policies through iterative self-exploration loops guided by automated GPT-4o trajectory evaluation.
ChatVLA-2 [238]	arXiv’25	Dynamic MoE SFT	Reasoning Token Projection	Disentangles control and understanding via dynamic MoE routing while injecting reasoning tokens into action layers.
LA4VLA [102]	arXiv’26	Vision-Agnostic Pretraining	None	Establishes robust language-action priors through vision-agnostic pretraining on atomic segments before visual grounding.
Latent Action Representations (Sec. 3.2.3)				
GenRL [117]	NeurIPS’24	Prompt-Guided Imagination	Latent Trajectory Matching	Aligns VLM embeddings with world model latents via noise-sampling to ground language prompts in physical dynamics.
LAWM [151]	arXiv’25	Self-Supervised Pretraining	Recurrent State-Space	Grounds latent action chunks in environment dynamics by jointly optimizing imitation learning with recurrent state-space models.
CLAP [209]	arXiv’26	Contrastive Pretraining	Rectified Flow Control	Disentangles visual dynamics from proprioceptive actions by contrasting state transitions against quantized robot action tokens.
Closed-Loop Control Mechanisms (Sec. 3.2.4)				
EMMA [198]	CVPR’24	DAGger-DPO	Parallel Text World	DAGger-DPO integrates retrospective LLM feedback to iteratively refine actions and minimize distribution shift.
COMBO [211]	ICLR’24	Tree Search Planning	Composable Video Diffusion	Beam-search tree integrates intent tracking and outcome evaluation to simulate cooperative trajectories from partial views.
UNeMo [68]	AAAI’25	Imitation Learning	CVAE State Inference	Hierarchical navigator fuses predicted visual features back into embeddings for fine-grained decision refinement.
Sim-to-Real Transfer Techniques (Sec. 3.2.5)				
Cosmos-Transfer1 [6]	arXiv’25	ControlNet Fusion	Spatiotemporal Conditionals	Spatiotemporal adaptive weighting fuses multimodal constraints to bridge visual domain gaps.
CLAP [209]	arXiv’26	Contrastive Pretraining	Rectified Flow Control	Contrastive alignment and Knowledge Matching regularization mitigate domain shift during embodiment adaptation.

dynamic loss weighting mechanisms further refine this process by scaling gradient contributions based on cross-modal consistency, effectively suppressing inertial guessing and prioritizing reliable sensory streams [178]. Such approaches integrate frame-aware spatiotemporal rewards to dampen coordinate oscillation, ensuring that the policy attends to valid perceptual evidence rather than static priors. To minimize cross-modal redundancy while maintaining task sufficiency, conditional information bottlenecks structurally separate verbalizable procedural skills from residual perceptual evidence [74]. This separation is achieved through a variational surrogate that maps text skill cards to a soft control prefix, yielding step consistency metrics that support a 39.18% success rate in complex manipulation tasks.

Exploration-Exploitation via Reward and Horizon Control. To mitigate inefficient search in high-dimensional action spaces, reward rescaling mechanisms

penalize redundant state transitions, compelling agents to prioritize high-efficacy actions over brute-force exploration [139]. This optimization strategy often couples with interaction frequency annealing, which dynamically constrains the reasoning horizon to distill multi-turn reactive loops into compact single-turn representations, thereby reducing episode length while maintaining proactive planning capabilities. Addressing the instability of token-level signals in sparse-reward settings, bi-level advantage estimation decouples credit assignment by integrating turn-level generalized advantage estimation with token-level propagation [159]. Furthermore, iterative exploration frameworks employ difficulty-guided sampling to selectively fine-tune policies on low-success trajectories identified through automated feedback, effectively breaking performance plateaus in open-world multimodal domains [62].

Explicit Reasoning and Failure Attribution. De-

detecting policy failures requires monitoring action-space anomalies and confidence thresholds to distinguish between perceptual neglect and linguistic misalignment [3]. Sharp entropy spikes in the action distribution or significant divergence from historical trajectory metrics serve as primary triggers for recovery protocols, prompting the system to re-evaluate the current state representation. This mechanism isolates irreversible reasoning errors, such as hallucinated interactions caused by scene complexity, by flagging instances where the policy cannot map observations to valid action primitives. By focusing on these internal failure signals, the framework identifies specific breakdown points in the decision chain through direct analysis of confidence degradation and action variance.

3.2.3 Latent Action Representations

Disentangling Abstract Actions via Information Constraints. Contrastive alignment objectives force visual state transitions to map onto a quantized action codebook, effectively filtering background noise from manipulation-relevant dynamics [209]. By employing a SigLIP-based cross-modal framework, these methods separate discrete reasoning from continuous visual entanglement through the generation of distinct robot action tokens. Complementing this spatial separation, a conditional information bottleneck structurally isolates verbalizable procedural skills from residual perceptual latents to minimize cross-modal redundancy [74]. This approach utilizes variational surrogate optimization and length-constrained prompting to ensure that text skill cards capture only task-sufficient information, leaving non-semantic details in the conditional latent vector. Such hierarchical decomposition assigns distinct architectural roles to symbolic planning and continuous flow control, preventing semantic interference during low-level execution [74, 209]. The resulting dual-formulation policies combine autoregressive next-token prediction with Rectified Flow-based Diffusion Transformers, using soft control prefix fusion to maintain task sufficiency without compromising motor precision.

Grounding via Dynamics Prediction. Bridging the semantic gap between high-level foundation embeddings and low-level physical execution requires aligning abstract prompts with generative world model latents without relying on explicit language annotations. Noise-sampling aligners address this by mapping vision-only backbone features, such as those from InternVideo2, into a shared latent space where cosine distance minimization enforces trajectory matching against predicted dynamics [117]. Extending this alignment, joint optimization frameworks integrate

recurrent state-space models to reconstruct future frame sequences from predicted latent action chunks, thereby acquiring label-free action priors through imitation objectives regularized by KL divergence [151]. This decoupling of policy learning from real-world data access enables agents to master novel tasks purely through imagination, minimizing the discrepancy between imagined dynamics and prompt-derived targets while discarding the need for supervised finetuning on embodiment-specific demonstrations [117, 151].

Hierarchical Composition and Granularity Control. Semantic forcing mechanisms align latent planning tokens with fine-grained event captions, effectively transferring explicit instructional grounding into implicit autonomous subgoal generation [142]. To sustain granular control over extended horizons, dual-memory architectures decouple causal visual dynamics from semantic progress tracking through retrieved history augmentation. This structural separation enables the system to maintain distinct long short-term event memories alongside causal short-term visual memories, utilizing attention pooling to synthesize global-history, local-active, and event-boundary views. Addressing the textual degradation rate inherent in one-dimensional symbolic skill serialization, visual skill priors externalize intermediate spatial states as rendered in-situ traces [188]. Such dynamic visual priors function as a form of visual working memory that anchors coordinates and mitigates omission or duplication errors during spatial bookkeeping. Furthermore, modality-specific binding protocols interleave declarative textual logic with reusable visual assets to resolve static boundary ambiguities, enabling adaptive granularity in complex manipulation sequences [188]. By compressing spatial topology and matching assets to specific modalities, these approaches facilitate the construction of interleaved skills that overcome the limitations of purely textual representations.

Regularization via Noise Injection. To enforce smooth latent interpolation and prevent mode collapse under partial observability, recent approaches inject structured stochasticity directly into the representation space. Specifically, applying a two-dimensional time-modality noise matrix introduces independent Gaussian corruption across both temporal and sensory dimensions, compelling the model to learn a robust joint distribution rather than memorizing discrete trajectory modes [75]. This fine-grained partial masking strategy optimizes a noise prediction objective, where minimizing the reconstruction error between the injected perturbation and the model’s estimate ensures consistent skill embeddings even

when input streams are incomplete or noisy.

3.2.4 Closed-Loop Control Mechanisms

Temporal Gating and Latency Management. Dynamic control loops modulate feedback frequency by gating high-latency reasoning modules only upon detecting specific uncertainty triggers or environmental ambiguities, effectively routing tasks between rapid heuristic processing and deliberate analysis [144, 235]. To maintain real-time responsiveness under strict temporal constraints, architectural designs decouple rapid perceptual inference from deliberate evidence accumulation cycles, utilizing switch adapters to bypass complex chain-of-evidence construction when region proposals or segmentation masks indicate low ambiguity. Furthermore, closed-loop agents optimize action selection intervals by modeling sequential information gathering as a higher-order decision process that balances observation cost against reasoning accuracy, often employing sufficiency judgment metrics to terminate iterative perception-reasoning-action cycles once visual token reduction via max pooling yields adequate state representation [235].

State Reconstruction and Predictive Correction.

To resolve partial observability, architectures reconstruct a unified global state from fragmented egocentric inputs by fusing point clouds and applying pixel inpainting to handle occlusions [211]. Hierarchical feedback loops further refine this process by injecting predicted future visual features directly into navigation embeddings, enabling cross-modal reasoning to correct trajectory deviations prior to execution [68]. Generative world models address decision myopia by evaluating diverse anchor-based trajectory hypotheses against simulated visual futures, thereby selecting robust action sequences while preventing mode collapse [118].

Retrospective Error Mitigation. Long-term memory buffers store execution critiques to enable iterative refinement of expert demonstrations, effectively minimizing distribution shift through mechanisms such as the DAgger-DPO framework where critic-generated feedback drives action label refinement [198]. Bidirectional data flows between semantic planners and physical simulators further ensure that logged execution failures directly update internal representations, creating a self-reflection loop that verifies physics compliance and aligns semantic intent with physical constraints [46]. By treating past execution errors as structured training signals rather than transient noise, these closed-loop architectures transform single-pass policies into adaptive systems capable of lifelong skill improvement via preference-

based alignment and trajectory hypothesis rejection [46, 198].

Runtime Graph Reconfiguration. To address fluctuating environmental conditions, adaptive policies dynamically reconfigure computational graph topology by allocating layer execution budgets according to real-time Quality of Information (QoI) and platform resource constraints [177]. This mechanism employs a differentiable NeuralSort controller to enable conditional layer skipping and feature complexity gating, effectively reducing FLOPs while maintaining decision fidelity under strict latency bounds. By continuously monitoring immediate sensory feedback quality, the system triggers topological adjustments that bypass non-essential processing nodes, ensuring stability during transient occlusions or sensor degradation without altering the underlying policy weights or relying on long-horizon curriculum stages [138].

3.2.5 Sim-to-Real Transfer Techniques

Domain Randomization via Physical Parameter Perturbation.

Bridging the simulation-to-reality gap requires exposing policies to diverse physical dynamics rather than relying on fixed generative backbones [6]. This method employs systematic perturbation of environmental and agent-specific parameters, such as friction coefficients, mass distributions, and sensor noise profiles, to prevent overfitting to specific simulated conditions. By training across a broad spectrum of randomized values, the policy learns invariant features that generalize to unseen real-world configurations without requiring explicit domain adaptation layers. Such stochastic variation ensures resilience against discrepancies in actuator response and terrain interaction, directly improving transfer success rates in complex scenarios where precise system identification is unavailable.

Contrastive Visual-Proprioceptive Alignment.

Mitigating the simulation-to-reality gap requires aligning visual state transitions with proprioceptive signals to suppress domain-specific artifacts while preserving task-critical dynamics. By employing a SigLIP objective, recent approaches couple visual dynamics with physical actions through a contrastive framework that maximizes mutual information between modalities, effectively isolating invariant features from embodiment discrepancies [209]. This alignment directly reduces the distributional shift between simulated training environments and physical deployment by forcing the policy to rely on cross-modal consistency rather than static background cues or simulator-specific textures. Consequently, policies generalize across varying robotic morphologies and

unseen real-world conditions without requiring additional data collection or retraining.

3.2.6 Discussion

Shifting from reactive policy learning to unified world-action modeling fundamentally alters the scalability landscape by enabling the utilization of massive action-free video datasets to learn physical priors [164, 227]. Bridging the sim-to-real gap increasingly relies on hierarchical latent representations and adaptive multimodal fusion rather than direct pixel-space mapping, allowing systems to handle morphological discrepancies through contrastive visual-proprioceptive alignment and quantized action codebooks [6, 142, 209]. Safety and reliability in open-world deployment necessitate active visual reasoning loops and dynamic cognitive switching, where uncertainty-triggered mechanisms address partial observability to prevent hallucinated actions [144, 235]. However, current architectural trade-offs reveal a critical tension between the computational latency of generative world models and the strict real-time requirements for safe closed-loop control, as cascaded architectures and event memory queues often strain GPU resources [6, 142, 164].

3.3 Cognitive Agentic Architectures

Cognitive agentic architectures integrate high-level reasoning with dynamic resource management to address complex, multi-step tasks. This section first examines frameworks where agents extend their capabilities through the dynamic selection and execution of external utilities in Sec. 3.3.1. We then analyze systems that maintain long-term episodic or semantic records to inform decision-making processes in Sec. 3.3.2, followed by an investigation into methods that ground generation within external knowledge bases to enhance factual accuracy in Sec. 3.3.3. The discussion subsequently expands to coordinated efforts among specialized entities in Sec. 3.3.4 before concluding with an exploration of algorithms designed to actively navigate solution spaces and reduce uncertainty in Sec. 3.3.5.

3.3.1 Tool-Augmented Reasoning

Semantic Tool Selection and Open-World Generalization. Decoupling semantic matching from fixed tool identifiers enables open-world generalization by transforming task descriptions into structured queries that align with vector space representations via cosine similarity [115]. To mitigate decision space complexity, strategic toolkit pre-selection establishes

high-level guidance through a global navigator module that identifies relevant subsets prior to iterative execution, thereby preserving context integrity [36]. Furthermore, adaptive coordination-path selection replaces rigid inference protocols with input-dependent routing mechanisms that predict trajectory success probabilities and calibrate query forms to generate compact reasoning paths [10].

Runtime Execution Failure Recovery. When tool invocation triggers runtime exceptions, embedding a feedback loop directly within the execution environment enables autonomous code revision by capturing error traces to guide iterative self-correction [36]. This mechanism analyzes the failure state to regenerate stateless snippets without requiring external intervention, focusing strictly on recovering from errors during dynamic operation rather than optimizing model parameters.

Resolution of Cross-Modal Conflicts. To enable coherent deduction across disparate modalities, recent frameworks prioritize the algorithmic weighting of heterogeneous signals over rigid format unification [119, 216]. This approach employs dual-agent nested loops to maintain a query-able communication channel, allowing the reasoning module to request targeted updates to the perceptual representation until specific confidence thresholds are satisfied. Systems address cross-modal conflicts arising from geometric reconstruction errors by dynamically weighing allocentric spatial priors against semantic cues within interwoven reasoning chains [134]. Such mechanisms utilize geometry-semantics alignment and modality-decoupled cue collection to resolve discrepancies in spatial parameters, ensuring that the integrated context remains robust against reconstruction artifacts.

Cost-Aware Invocation Optimization. To mitigate the computational overhead of external tool usage, reinforcement learning policies are trained to balance reasoning accuracy against invocation costs by penalizing unnecessary calls [122, 206]. These approaches employ composite reward functions within frameworks like Group Relative Policy Optimization (GRPO) to enforce sparse intermediate steps and promote selective activation only when strictly required. Furthermore, generative imagination tools are restricted to scenarios where perceptual ambiguity exceeds a defined threshold, preventing redundant visual synthesis during occlusion resolution or cognitive mapping tasks [206]. By gating diffusion-based outpainting and open-vocabulary human-object interaction analysis behind efficiency constraints, these systems avoid the latency penalties associated with unconditional generation.

Table 3 Taxonomy of representative methods in **Cognitive Agentic Architectures**. This table presents a taxonomy of cognitive agentic architectures, shows distinct reasoning cores and memory substrates, and highlights the shift toward structured graph-based knowledge integration.

Method	Venue	Reasoning Core	Memory Substrate	Highlight
Tool-Augmented Reasoning (Sec. 3.3.1)				
ToolScope [36]	ACL’25	Modular Tool Execution	Queryable Visual Regions	Global Navigator selects tool subsets while Perceive treats images as queryable memory for dynamic re-attention.
World2Mind [134]	arXiv’26	Geometry-Semantic Alignment	Spatial Cognitive Maps	Allocentric-Spatial Tree structures spatial cues to enable training-free, modality-decoupled tool invocation and conflict resolution.
RaTA-Tool [115]	arXiv’26	Retrieve-then-Generate	Vector Tool Embeddings	Direct Preference Optimization aligns structured task descriptions with tool embeddings to maximize open-world retrieval accuracy.
Memory-Augmented Planning (Sec. 3.3.2)				
JARVIS-1 [171]	TPAMI’23	Hierarchical Planner	Multimodal Trajectory Logs	Stores multimodal task-situation-plan tuples for backward-decomposed query retrieval to inform situation-aware planning.
WKM [127]	NeurIPS’24	Dual-Model Planning	Action-State Triplets	Indexes action-state-action triplets to retrieve local constraints that guide policy via weighted summation during inference.
AStar [182]	AAAI’25	Pattern-Guided MCTS	Abstract Thought Cards	Retrieves abstract thought cards by matching problem complexity and semantics to instantiate external reasoning guidelines.
Retrieval-Augmented Generation (Sec. 3.3.3)				
VaLiK [106]	ICCV’25	Cascade Verification	Purified Visual Triplets	Sliding window cross-modal similarity verification prunes hallucinated tokens before constructing retrieval-augmented multimodal knowledge graphs.
MMGraphRAC [156]	arXiv’25	Cross-Modal Linking	Unified Scene Graphs	Spectral clustering aligns visual and textual entities for precise cross-modal linking and reasoning path retrieval within unified graphs.
RAG-Anything [55]	arXiv’25	Dual-Graph Navigation	Heterogeneous Entity Graphs	Dual-graph construction merges structural navigation with dense semantic matching to retrieve dispersed evidence across modalities effectively.
MG ² -RAG [34]	arXiv’26	Graph Propagation	Multi-Granularity Nodes	Entity-driven visual grounding unifies nodes for multi-granularity retrieval via Personalized PageRank propagation across heterogeneous topologies.
MAGE-RAG [240]	arXiv’26	Budgeted Graph Traversal	Evidence Subgraphs	Budgeted online controller dynamically constructs evidence subgraphs through iterative activation and pruning actions for efficient multimodal retrieval.
Multi-Agent Collaboration (Sec. 3.3.4)				
COMBO [211]	ICLR’24	Generative Tree Search	Video Diffusion Latents	Factorizes joint actions via beam-search tree with intent tracking for composable multi-agent video generation.
MindSearch [28]	ICLR’24	Code-Driven DAG	Live Web Index	Models reasoning as executable DAGs to dispatch parallel WebSearcher agents for hierarchical query decomposition.
MDocAgent [58]	arXiv’25	Parallel Dual-Stream	Segmented Document Blocks	Employs a critical agent to guide specialized modality workers, resolving cross-modal conflicts through structured synthesis.
MATHAGENT [193]	ACL’25	Sequential Validation	Educational Data Streams	Decomposes error detection into sequential validators and interpreters to localize multimodal misconceptions via adaptive conversion.
Kimi K2.5 Agent Swarm [11]	arXiv’26	Parallel Agent Swarm	Trillion-Token Corpus	Deploys a trainable orchestrator to instantiate frozen sub-agents dynamically, optimizing parallel execution without co-training.
Search and Exploration Strategies (Sec. 3.3.5)				
MindSearch [28]	ICLR’24	Code-Driven DAG	Live Web Index	Models reasoning as DAG construction via code, dispatching parallel agents for hierarchical retrieval.
Scalable Bayesian ToM [210]	ICML’25	Bayesian Inverse Planning	LM Parameter Weights	Decomposes reasoning into stepwise Bayesian updates, steering large model distributions via weak-to-strong control.
LIRAS [203]	EMNLP’25	Neuro-Symbolic POMDP	Symbolic State Sequences	Synthesizes PDDL models on-the-fly to execute Bayesian inverse planning over latent mental states.
GraphVista [57]	ACL’25	Hierarchical Routing	Topological Graph Tiers	Routes tasks dynamically between textual retrieval and visual graph thoughts using topological centrality tiers.
MMhops-R1 [214]	AAAI’25	RL Policy Selection	Wiki Knowledge Base	Optimizes dynamic multi-modal action selection via reinforcement learning with composite reward gating.

Syntax and Schema Standardization. Achieving interoperability across heterogeneous visual, textual, and audio domains requires enforcing a unified structured JSON schema that normalizes diverse inputs into a consistent syntactic format [115, 119]. This architectural standard transforms multimodal

queries and tool metadata into compact, schema-compliant objects, enabling direct data exchange without modality-specific adapters. By encoding visual cues alongside textual descriptions into this common structure, systems decouple task semantics from input modalities, allowing agents to process

information based on standardized field definitions rather than raw sensory formats.

3.3.2 Memory-Augmented Planning

Structural Indexing and Retrieval Mechanisms.

Moving beyond linear trajectory storage, contemporary architectures decompose experience into atomic state-action units indexed by compositional embeddings to isolate high-quality decision knowledge [163]. This abstraction facilitates hindsight reasoning and enables multi-viewpoint indexing through iterative deep-and-wide search strategies that prioritize specific interaction granularities. To capture complex logical dependencies, dynamic directed acyclic graph topologies have replaced sequential history buffers, allocating visual token density based on node energy scores and topological centrality [161]. Such graph-modulated encoding allows the system to track dependency chains rather than mere temporal order, optimizing resource distribution across the memory structure. At a higher level of formalization, episodic knowledge graphs decouple raw signal segmentation from semantic interpretation by utilizing ontology-mapped URIs [135]. By representing signal streams in formats such as JSON-LD and employing RDF-based structures with specialized ontologies, these systems support co-existing alternative interpretations while maintaining rigorous uncertainty provenance.

Consistency Maintenance and Obsolescence Filtering.

Preventing decision fidelity degradation in dynamic environments requires agents to actively filter obsolete state information rather than relying on append-only memory accumulation [105, 171]. To mitigate attention dilution caused by expanding context windows, adaptive history compression mechanisms convert stale textual observations into compact visual tokens, preserving raw evidence fidelity while preventing lost-in-the-middle failures [108]. Interactive planning loops further enforce consistency by integrating proactive self-checks and reactive self-explanations that validate stored memories against current multimodal states prior to execution [171]. Benchmarking frameworks support these architectural choices by decomposing the memory lifecycle into distinct write, maintain, retrieve, and use stages, thereby isolating specific failure modes within the maintenance phase [105].

Fidelity-Efficiency Trade-offs. To resolve the tension between high-fidelity spatial state tracking and computational overhead, recent architectures externalize working memory into rendered visual traces rather than expanding textual context windows [188]. This approach mitigates textual degradation rates

by employing in-situ visual traces for spatial book-keeping, which significantly reduces omission errors through coordinate anchoring and dynamic priors. Alternatively, feature-space state transition prediction offers a computationally efficient path that avoids pixel-level reconstruction while maintaining sufficient fidelity for long-sequence consistency [50]. By utilizing progressive state transition training and cross-attention grounding within modules such as ContextReflector, these systems achieve modality reconstruction latency improvements and enhanced few-shot adaptation efficiency without the burden of pixel-space diffusion.

Retrieved Experience in Adaptation and Generalization.

Few-shot adaptation emerges when agents retrieve abstract reasoning templates rather than raw trajectories, allowing lightweight libraries of distilled patterns to guide implicit model capabilities without retraining [182]. Policy generalization further improves by decoupling global task priors from dynamic state constraints, where self-synthesized instance-level knowledge imposes retrievable action-state restrictions on local planning [127]. Cross-modal transfer relies on constructing allocentric spatial representations that transform egocentric sensory inputs into structured topological maps, enabling text-only backbones to execute complex geometric reasoning [134].

3.3.3 Retrieval-Augmented Generation

Graph-Structured Relevance and Noise Suppression.

Moving beyond flat vector retrieval, structured graph topologies enforce logical connectivity to suppress hallucinations during context grounding [34, 156]. These architectures employ spectral clustering and cross-modal entity linking to fuse atomic visual evidence with scene graphs, utilizing Personalized PageRank for modality-preserving fusion. To prevent structural fragmentation in long documents, hierarchical and dual-graph designs isolate fine-grained visual anchors from semantic relations through page-level subgraphs and GME embedding [55, 65]. Such systems integrate layout parsing with multi-signal fusion scoring to maintain coherence across complex entity-relation extractions.

Multimodal Evidence Integration in Generation.

Advanced frameworks integrate pre-retrieved visual evidence directly into the text generation stream to ground outputs without active search loops [29, 160]. These mechanisms align high-resolution image features with semantic requirements by fusing coarse-to-fine visual representations within the decoder context, ensuring that fine-grained details inform token

prediction. This approach utilizes static visual snapshot representations and temporal decay factors to preserve layout semantics while weighting the contribution of atomic evidence to the final output [125].

Direct Multimodal Projection in Generation. Heterogeneous visual and tabular inputs are projected directly into the generator’s latent space, bypassing intermediate text serialization to preserve modality-specific fidelity [16]. Specialized cross-attention mechanisms align high-resolution image features and structured table embeddings with semantic query vectors within the decoder context, ensuring that raw non-textual evidence informs token prediction without deterministic command parsing [119]. Evidence-grounded recaptioning modules function here as auxiliary constraints that explicitly disentangle identity preservation from scene composition logic during the attention weighting phase, rather than serving as pre-processing steps [25]. Finally, annotation-free cascade architectures utilize cross-modal similarity verification to dynamically adjust attention scores over hierarchical visual descriptions, enabling the model to link entities with their visual counterparts through continuous embedding alignment instead of static triplet extraction [106].

Latency Optimization via Sparse Graph Pooling. To mitigate the computational overhead of exhaustive semantic matching, retrieval-based graph pooling strategies replace dense similarity searches with lightweight extraction mechanisms that operate directly on sparse knowledge structures. By employing path searching and entity matching pooling within cross-modal vision-text graphs, these approaches drastically reduce token consumption, achieving savings of up to 94.9 percent while preserving essential structural information [47]. This static efficiency relies on architectural constraints within the retrieval module to limit context expansion, ensuring that trajectory management remains within predefined length limits without sacrificing the relevance of retrieved evidence through selective subgraph isolation [240].

Conflict Resolution and Consistency Verification. To resolve semantic ambiguities during multi-source grounding, hierarchical frameworks align extracted entity-event-scene tuples with structured knowledge graph nodes using weighted action and description similarity metrics [92]. Parallel retrieval streams address conflicting interpretations by executing granular consistency checks that synchronize text and visual data within collaborative topologies, replacing the need for autonomous decision-making units [58]. Preventing cognitive drift during self-evolution

requires iterative truth anchor expansion, where verified ground-truth samples retrieved via semantic keywords replace low-quality synthetic supervision [169]. Furthermore, cross-modal consistency is enforced by retrieval checkers that filter semantically misaligned triplets using encoder similarity thresholds before augmenting generation with concept-centric knowledge [126].

3.3.4 Multi-Agent Collaboration

Topological Task Decomposition and Parallel Execution Graphs. Complex reasoning problems are structurally reformulated as executable Directed Acyclic Graphs (DAGs), where atomic sub-questions function as dynamic nodes generated through code synthesis rather than static prompting [28]. In multimodal contexts, this architecture employs parameter-sharing virtual subagents to isolate specific visual operations within a causal ordering constraint, effectively decoupling perception modules from language coordination to reduce modality switching costs [197]. Distinct from parallel graph structures, specialized pipelines enforce strict sequential dependency chains that utilize binary consistency filtering and adaptive semantic conversion to localize errors before integrative analysis occurs [193]. These divergent topological strategies collectively shift the computational burden from monolithic generation to structured, verifiable execution paths.

Grounded Communication via Latent Prediction. Semantic alignment between linguistic signals and unobserved visual states is enforced by conditioning latent trajectory predictions on incoming messages, compelling agents to compress visual semantics into a shared recurrent state-space model [4, 32]. This message-conditioned prediction framework utilizes auxiliary reconstruction losses to decouple visual content from communication protocols, ensuring that transmitted tokens reflect specific environmental dynamics rather than generic descriptions. To operationalize this alignment, hierarchical attention mechanisms fuse asymmetric multimodal observations into a unified belief state prior to policy execution, facilitating centralized training with decentralized inference [4]. Such fusion architectures maintain robustness against sensor dropout by optimizing a compact latent space that integrates cross-modal features without relying on raw data transmission.

Conflict Resolution through Weighted Aggregation and Visual Flow. To mitigate error accumulation in multi-step visual reasoning, frameworks construct a Multi-View Knowledge Base where hypothesis-confidence pairs undergo score-

based weighted voting, effectively emulating top-down reasoning by linking issues to answers and injecting zero-shot common sense [170]. Addressing the propagation of hallucinations across agent turns, alternative architectures suppress snowballing effects by relaying unimodal vision tokens directly through middle layers while dynamically reallocating attention weights via temperature scaling and mask matrix application to amplify unimodal peaks [205]. In document understanding scenarios involving cross-modal interpretive conflicts, a critical agent resolves discrepancies by extracting granular information guides that synchronize specialized text and image agent outputs within a five-role collaborative topology prior to final synthesis [58].

Scalability via Composition and Compression.

Factorizing joint action spaces into composable text components allows world models to generalize across arbitrary agent counts without retraining, as demonstrated by compositional video diffusion approaches that integrate partial view inpainting with tree search [211]. To address context expansion limits, alternative frameworks convert linear debate histories into compact visual representations, theoretically guaranteeing information recovery through agent diversity despite aggressive token compression [181]. Parallel execution architectures further decouple orchestrator optimization from frozen sub-agent policies, transforming sequential latency bottlenecks into concurrent processing streams via dynamic instance instantiation and outcome-based reward shaping [11].

3.3.5 Search and Exploration Strategies

Quantifying Information Value. Post-hoc counterfactual analysis transforms sparse global outcomes into dense turn-level rewards by estimating the marginal utility of atomic evidence units through success probability differentials and temporal decay factors [125]. Distinct from retrospective attribution, explicit modeling of information sufficiency within higher-order Markov Decision Processes enables agents to predict action-conditioned information gain prior to execution, utilizing Chain-of-Thought supervision to optimize sequential visual integration under occlusion [235]. Complementing these structural approaches, weak-to-strong control mechanisms steer large language model priors toward specific belief update behaviors via inference-time logit ratio adjustments, effectively scaling multimodal Theory-of-Mind planning without requiring additional training data [210].

Scalable Search Algorithms for High-Dimensional Multimodal State Spaces. To

mitigate state blindness in token-heavy visual environments, dynamic graph topologies replace linear interaction histories by capturing logical dependencies between agent states through directed acyclic graph construction [161]. This architectural shift utilizes node energy scores and topological centrality within recursive feedback loops to optimize visual token density allocation, ensuring critical information persists beyond immediate context windows. Hierarchical partitioning further enhances scalability by routing queries to specific topological tiers—Core, Backbone, and Peripheral—based on modality-specific reasoning requirements [57]. By applying PageRank segmentation and Betweenness Centrality metrics, these systems extract relevant visual subgraphs and direct tasks to appropriate reasoning layers without exhaustive traversal. For complex query resolution, explicit decomposition into executable code-generated DAGs facilitates parallel search execution across independent threads [28]. This approach synthesizes atomic sub-questions into Python-based structures that dispatch concurrent retrieval processes, aggregating hierarchical results with global context injection to effectively manage information noise.

Adaptive Cost Regulation. To sustain optimal gradient utility during reinforcement learning, adaptive curriculum sampling mechanisms dynamically modulate task difficulty distributions based on online performance metrics such as pass@ k monitoring and ratio-based Exponential Moving Averages [64]. These approaches align challenge levels with agent capability to prevent reward hacking, though they remain sensitive to cold-start initialization conditions. Distinct from curricular adjustments, stateful agentic controllers enforce strict computational budgets by iteratively activating or pruning evidence nodes within multigranular graph structures according to marginal utility estimates [240]. This selective evidence purity optimization mitigates context explosion while introducing latency overhead tradeoffs inherent to query-time budget constraints. Furthermore, reinforcement learning policies can autonomously determine variable-length reasoning trajectories by optimizing a composite reward function that explicitly trades off correctness against cumulative interaction costs [214]. Such dynamic trajectory termination drives interaction rounds toward plateauing points, yet the efficacy of this strategy scales directly with retriever quality and incurs non-trivial computational overhead.

Escaping Local Optima via Stochastic Frontiers. Iterative discovery of failing frontiers employs stochastic rollouts to actively isolate model

failure modes, subsequently triggering the retrieval of verified ground-truth anchors that replace hallucinated synthetic data [169]. This mechanism mitigates cognitive drift by expanding the training distribution with truth-aligned samples rather than reinforcing erroneous generative patterns. Complementing this approach, scalable synthesis of high-uncertainty tasks utilizes random-walk graph sampling combined with information obfuscation to force agents away from low-uncertainty trajectories [90]. By prioritizing structural complexity and non-linear reasoning patterns over simple solution paths, such methods ensure that optimization proceeds through regions of maximal informational gain.

Pruning Search Spaces via Priors and Structured World Models. Soft logic predicates derived from large language model priors constrain frontier selection by balancing geometric distance with semantic co-occurrence probabilities, effectively mitigating distribution shifts in zero-shot navigation through probabilistic soft logic and one-hot constraint solving [232]. Distinct from purely geometric approaches, typed knowledge graphs unify data synthesis and environment simulation to enable step-level credit assignment via hop-anchored policy optimization, utilizing gold entity anchoring and anti-shortcut filtering to resolve structural disconnects without external reward models [82]. Furthermore, on-the-fly synthesis of symbolic Partially Observable Markov Decision Processes allows neural perception modules to perform Bayesian inverse planning over latent mental states, where generated PDDL models facilitate graded social inference through sequential inverse agent modeling [203].

3.3.6 Discussion

Functional autonomy in current cognitive architectures arises not from monolithic reasoning but through uncertainty-aware delegation and selective tool invocation, where agents dynamically route tasks based on risk metrics and evidence horizons [122, 134, 217]. Scaling these systems to open-ended environments necessitates structured memory topologies and iterative search strategies that prevent context window exhaustion by employing dynamic DAGs, energy-based token allocation, and unified text space representations [16, 119, 161, 220]. Despite these advances, planning bottlenecks persist due to dependencies on synthetic data quality, latent imagination limits, and the computational overhead inherent in graph traversal or multi-stage pipelines [4, 90, 122, 217]. Information obfuscation and cascading errors in reconstruction fidelity further constrain performance, revealing a critical gap between theoretical architectural flexibility and the practical latency costs of

maintaining physical fidelity in dynamic settings.

3.4 Unified Multimodal Representation

This section establishes the foundational architectures required to map heterogeneous data sources into coherent, shared embedding spaces for effective downstream reasoning and generation. We first examine methods designed to project distinct modalities into a common semantic manifold, enabling direct comparison and fusion as detailed in Sec. 3.4.1. The discussion then extends to the incorporation of structured symbolic knowledge, which enhances interpretability and logical consistency within multimodal systems (Sec. 3.4.2). Subsequently, we analyze techniques for merging features at the latent level to capture complementary information through Sec. 3.4.3, followed by an investigation into modality-agnostic encoders that utilize a single shared backbone for improved parameter efficiency (Sec. 3.4.4). Finally, the section concludes by exploring disentangled representation learning strategies that separate content, style, and modality-specific factors to maximize generalization and controllability (Sec. 3.4.5).

3.4.1 Cross-Modal Alignment Spaces

Contrastive Objective Functions for Shared Semantic Mapping. Resolving the theoretical conflict where increasing modality counts degrade mapping quality requires decoupling alignment and uniformity objectives through mechanisms such as anchor-based alignment, intra-modality Gaussian kernel density estimation, and global Hölder divergence minimization [202]. Dynamic centroid generation further eliminates modality-specific bias by replacing fixed reference anchors with batch-level aggregation to simultaneously optimize intra- and inter-modal mutual information via symmetrized InfoNCE loss [79]. To prevent encoders from relying on dominant single streams, constructing disturbed negative tuples with partially mismatched modalities forces the learning of synergistic correlations through bilevel optimization and k-disturbed strategies [109]. Direct joint alignment of raw geometric data with language and image spaces is achieved without intermediate depth projections by utilizing triplet proxy collections that enforce joint semantic-instance objectives [208]. Finally, integrating fine-coarse masked contrastive learning with modality-agnostic jigsaw puzzles establishes robust unified boundaries for open-set generalization through sample-specific masking and discrete code-word permutation [71].

Architectural Mechanisms for Cross-Modal Attention and Fusion. Spatially weighted cross-attention

Table 4 Taxonomy of representative methods in **Unified Multimodal Representation**. This table compares unified multimodal methods by alignment strategy and disentanglement mechanism, showing architectural diversity and highlighting the prevalence of contrastive objectives.

Method	Venue	Alignment Strategy	Disentanglement Mechanism	Highlight
Cross-Modal Alignment Spaces (Sec. 3.4.1)				
PRH [76]	ICML’24	Statistical Convergence	Theoretical Factorization	Formalizes cross-modal alignment as convergence to Point-wise Mutual Information kernels of latent reality distributions.
DMRNet [173]	ECCV’24	Probabilistic Embedding	Stochastic Sampling	Models embeddings as Gaussian distributions, decoupling stochastic training samples from deterministic inference means to relax constraints.
Open-World Classification [30]	ICCV’25	Generative Matching	Metric-Based Filtering	Proposes four metrics quantifying semantic and textual alignment nuances for open-world classification evaluation.
MICU [71]	ICCV’25	Unified Jigsaw	Masked Contrast	Integrates Fine-Coarse Masked InfoNCE and modality-agnostic Jigsaw Puzzles to enforce holistic and temporal alignment.
TLV-CoRe [236]	AAAI’25	Tri-Modal Contrast	Adversarial Decoupling	Projects heterogeneous sensor embeddings into shared space via Unified Bridging Adapter with adversarial sensor-invariance objectives.
UniAlign [202]	arXiv’26	Decoupled Contrast	Anchor-Based Alignment	Decouples alignment and uniformity using anchor-based strategies to eliminate non-collinear positive pull cancellation.
Knowledge Graph Integration (Sec. 3.4.2)				
VaLiK [106]	ICCV’25	Cascade Verification	Similarity Pruning	Cascade VLMs generate triplets while cross-modal similarity verification prunes hallucinated relations.
SSU [145]	AAAI’25	Graph Fusion	Semantic Anchoring	Text-guided dynamic graphs and semantic anchors unify heterogeneous modalities via contrastive objectives.
MegaRAG [65]	Volume’25	Dual-Level Retrieval	Iterative Refinement	Subgraph-guided refinement retrieves missing cross-modal links to correct initial entity-relation extractions.
Semantic Latent Fusion (Sec. 3.4.3)				
Any2Seg [230]	ECCV’24	Knowledge Distillation	Correlation Reweighting	Distills inter-modal correlations via language guidance and selects fine-grained features using cross-modal similarity maps.
MAPI-GNN [128]	AAAI’25	Hierarchical Fusion	Perturbation Analysis	Constructs dynamic semantic activation graphs to hierarchically fuse intra-sample interactions and inter-sample patient relationships.
LatentUM [83]	arXiv’26	Latent Quantization	Expert Specialization	Unifies understanding and generation in a shared discrete latent space using mixture-of-modal experts with decoupled feedforward networks.
Modality-Agnostic Encoders (Sec. 3.4.4)				
OmniVec [141]	WACV’24	Shared Backbone	Meta-Token Projection	Projects inputs to shared space via meta-token conditioning, processing unpaired multimodal data through a common transformer bottleneck.
Hulk [168]	TPAMI’23	Modality Translation	Learnable Indicators	Unifies diverse tasks as modality translation using learnable indicators within a single shared encoder-decoder architecture without task-specific heads.
Any2Seg [230]	ECCV’24	Knowledge Distillation	Correlation Reweighting	Distills semantic correlations from teachers and fuses features via similarity indexing to generate robust modality-agnostic representations for segmentation.
MICU [71]	ICCV’25	Unified Jigsaw	Masked Contrast	Aligns features via masked contrastive loss and performs modality-agnostic permutation learning on unified discrete codeword streams.
SATA [215]	AAAI’25	Joint Task Training	Decoupled MoE	Separates common and specific knowledge via decoupled mixture-of-expert modules within a transformer to unify diverse tracking subtasks.
Disentangled Representation Learning (Sec. 3.4.5)				
SRCID [70]	ICASSP’25	Cross-Modal CPC	Residual Separation	Hierarchical encoder separates general and specific features via mutual information minimization and semantic residual extraction.
URMMDG [72]	ICCV’25	Contrastive Learning	MI Minimization	Contrastive learning disentangles category-general and modality-specific components to enable synchronized multimodal augmentation.
DMIL [199]	arXiv’26	Variational Decomposition	Component Factorization	Variational decomposition factors representations into redundancy, uniqueness, and synergy components using information-theoretic bounds.

enforces fine-grained binding between textual semantics and specific visual trajectory regions, utilizing Gaussian heatmaps and bounding box biasing to resolve multi-agent alignment ambiguities [157]. Decoupled training stages isolate behavioral alignment

from latent space optimization, where bidirectional Transformers model cross-modal dependencies more effectively than unidirectional approaches through progressive text conditioning replacement [186]. To address temporal aggregation, graph construction via

capsule network dynamic routing replaces recurrent mechanisms, fusing long-range dependencies without the information loss inherent in pooling operations [180]. Furthermore, dynamic text-guided graph construction coupled with a global semantic anchor hub unifies heterogeneous modality-specific structures, enforcing structural robustness through multi-view contrastive objectives and adaptive sparsification [145].

Regularization Strategies to Prevent Modality Collapse and Dominance. Explicit representational decoupling via orthogonality constraints mitigates feature conflict when aligning heterogeneous foundation model experts within a shared diffusion latent space, as demonstrated by M2-REPA which employs Centered Kernel Alignment (CKA) regularization alongside modality-specific MLP projectors to enforce cosine similarity loss and resolve multi-expert interference [185]. To counteract optimization laziness where stronger modalities suppress weaker signals, uni-modal feature distillation from frozen teachers enforces discriminative representation learning prior to fusion; the Uni-Modal Teacher (UMT) framework utilizes L2 distillation loss with pre-trained encoder freezing to facilitate task-adaptive ensemble selection and feature-level regularization [42, 43]. Probabilistic embedding sampling further relaxes rigid intra-class directional constraints to preserve modality-specific diversity during joint optimization of imbalanced data streams, with DMRNet modeling features as multivariate Gaussian distributions to implement stochastic training-inference decoupling and hard combination regularizers that block detrimental gradient directions based on variance magnitude correlations [173]. Finally, adversarial sensor-identity removal eliminates domain-specific artifacts to stabilize gradient updates and ensure balanced contribution across non-standardized tactile inputs, exemplified by TLV-CoRe which integrates a Sensor-Aware Modulator (SAM) for mutual information minimization and adversarial decoupling within tri-modal contrastive alignment routing weights [236].

Robustness to Missing Modalities via Imputation and Distributional Alignment. Handling incomplete input streams requires aligning distinct modality distributions within a unified concept manifold, where missing features are approximated through nearest-neighbor retrieval of aligned local concepts [41]. To mitigate projection domain shift during generative feature synthesis, recent approaches integrate Wasserstein distance directly into reconstruction losses, optimizing a single encoder-decoder architecture to harmonize disparate statistical moments [13]. Complementary strategies employ symmetric KL-divergence constraints between full and partial

posteriors within hierarchical variational frameworks, preventing expert overconfidence when specific sensors drop out [155]. Furthermore, semi-supervised cycle-consistency objectives enforce invertible mappings in Global Workspace architectures, enabling effective cross-modal translation with significantly reduced paired data requirements by assuming bijective relationships between frozen specialist encoders [38].

Quantitative Metrics for Alignment Quality and Uniformity. To move beyond rigid string matching, open-world classification protocols now employ multi-metric semantic inclusion scores that quantify alignment granularity and synonym tolerance [30]. These frameworks integrate text and Llama-based inclusion measures alongside concept similarity metrics to construct error taxonomy quadrants, effectively discriminating between specificity deficits and hallucinations. Complementing these empirical tools, theoretical analyses link cross-modal alignment quality directly to individual model competence through convergence toward shared statistical kernels of reality [76]. Grounded in the Platonic Representation Hypothesis, this perspective utilizes mutual nearest-neighbor metrics and PMI kernel approximations to demonstrate linear scaling trends where representational variance reduces as models align with a common underlying structure.

3.4.2 Knowledge Graph Integration

Visual Entity Grounding and Symbolic Node Linking. Mapping detected objects to predefined lexical hierarchies or dynamic synsets establishes the symbolic anchors required for structured reasoning, often utilizing WordNet hypernym relations alongside few-shot classifiers and segmentation masks to resolve open-vocabulary entities [34, 133]. Once grounded, cross-modal entity linking fuses independent visual scene graphs with textual knowledge nodes by applying spectral clustering and cosine similarity thresholds to align scene graph head-relation-tail triplets within sliding verification windows [106, 156]. To maintain graph integrity, noise pruning mechanisms enforce cross-modal consistency checks that filter hallucinated relations through concreteness scoring, temporal coherency constraints, and token-level removal of spurious OCR artifacts before final insertion [106, 133]. Advanced architectures further bypass intermediate text translation by constructing unified multimodal nodes that directly associate visual regions with textual entities, enabling modality-preserving fusion and atomic evidence retention within a single heterogeneous topology supported by Personalized PageRank propagation [34, 156].

Heterogeneous Graph Topologies. Bridging embodied sensorimotor vectors with amodal symbolic embeddings requires dynamic message passing across bipartite item-modality structures, often optimized via RotatE loss in complex space to align disparate feature distributions [21, 85]. Rather than relying on static connectivity, adaptive graph topologies emerge by projecting features into semantic subspaces where perturbation analysis identifies salient nodes through multi-dimensional discriminators and text-guided cross-attention [128, 145]. Global structural coherence is further enforced by injecting shared semantic anchors or hierarchical relations into modality-specific graphs, utilizing planar graph encoders and inter-sample modeling to unify views under multi-view contrastive objectives. To address scalability amidst arbitrary modality counts, undirected graph topologies are pruned using minimum spanning trees derived from Kruskal’s algorithm, which removes redundant edges based on correlation weights while maintaining robustness to missing data [73].

Dynamic Graph Evolution and Continual Updates.

To balance stability and plasticity during incremental learning, recent approaches modulate inter-modality gradient flows using adaptive coefficients and asymmetric attention distillation losses, thereby preventing catastrophic forgetting while integrating new multimodal data [26]. Contextual relevance drives structural expansion in alternative frameworks, where systems extract specific subgraphs from resources like ConceptNet and employ sequential evolution strategies to align symbolic structures with immediate environmental states [120]. Neuro-symbolic loops further refine these representations by coupling inductive inference from large language models with deductive validation via Answer Set Programming, ensuring consistent common ground as procedural knowledge graphs update through sensorimotor experience [88]. Finally, efficient state propagation is achieved by modeling transitions as graph queries, where action nodes trigger multi-hop aggregation under homophilous constraints to reduce token costs while updating multimodal state structures [45].

Probabilistic Edge Completion and Uncertainty

Quantification. Distinct from deterministic pruning strategies, recent frameworks address structural incompleteness by modeling relation existence as a latent variable subject to variational inference. These methods estimate posterior distributions over graph topology to compute probabilistic bounds for missing links, thereby inferring edges based on statistical likelihood rather than fixed similarity thresholds. By quantifying epistemic uncertainty in cross-modal connections, the architecture assigns continuous confi-

dence scores that preserve ambiguous entities during initial aggregation phases. This formulation defers binary inclusion decisions until global consistency constraints resolve local ambiguities, allowing the system to maintain topological flexibility under data scarcity [73, 106, 166].

Symbolic Primitives and Logical Constraints. By encoding explicit ontological constraints and logical rules directly within the graph topology, recent frameworks achieve composable counterfactual reasoning that frozen implicit neural weights cannot support [97]. These systems utilize deterministic rule evaluation and precise symbolic patch updates to yield high decomposability scores, ensuring that inference steps remain traceable to specific structural axioms. Extending beyond isolated chunk processing, retrieving structured subgraphs alongside raw multimodal pages injects global structural context that mitigates modality bias [34, 65]. Such approaches employ dual-level retrieval strategies to aggregate heterogeneous topology, significantly improving holistic cross-modal reasoning performance without relying on iterative propagation mechanics. Furthermore, decoupling textual and visual reasoning stages within these graph-augmented pipelines prevents dominance bias and enhances the logical coherence of generated explanations [23, 65]. The integration of dependency trees and formal logical operators within the graph structure enables the detection of inter- and intra-modal inconsistencies, thereby grounding interpretability in verifiable symbolic contradictions rather than statistical correlations.

3.4.3 Semantic Latent Fusion

Depth-Adaptive Fusion Strategies and Hierarchical Integration.

Moving beyond static topologies, hierarchical graph construction dynamically partitions latent features into semantically distinct subspaces to model patient-specific pathological dependencies [128]. This approach utilizes intra-sample graph attention networks alongside inter-sample graph convolutional networks, optimized via a composite loss that enforces consistency across M semantic subspaces. In single-stream early fusion architectures, modality-aware projectors enforce intra-modal consistency constraints through variance-covariance regularization, preventing dominant modalities from suppressing complementary signals during unified representation learning [146]. Such pipelines further benefit from language-guided distillation, where multi-view language model teachers transfer inter-modal correlation knowledge to enable fine-grained feature selection via similarity-based reweighting [230]. Extending these principles to decision-making contexts, multimodal

world models employ recurrent state-space mechanisms to aggregate asymmetric sensory observations into a compact latent belief state, facilitating sample-efficient policy imagination through hierarchical attention [4].

Attention-Based Modality Re-weighting and Bias Mitigation. Dynamic centroid generation replaces fixed modality anchors to eliminate selection bias, enforcing symmetric intra- and inter-modal alignment through adaptive re-weighting mechanisms [79]. To counteract ‘Modality Laziness’ where dominant signals suppress weaker ones, uni-modal feature distillation regularizers enforce independent representation learning prior to fusion, often utilizing teacher-student frameworks with late-fusion constraints [43]. Learnable query embeddings further refine this balance by progressively internalizing control signals, allowing weighted semantic averaging to maintain perceptual fidelity alongside abstract reasoning during latent space alignment [186]. Finally, multi-head fusion modules restore fine-grained complementary information attenuated by attention dilution in deep layers, employing dynamic consistency and subspace gating to isolate patch-level features [183].

Generative Decoding and Cross-Modal Reconstruction. Unified encoder-decoder frameworks employ distributional alignment losses, such as Wasserstein distance, to reconstruct missing modal features and mitigate projection domain shift in zero-shot scenarios [13]. Masked cross-modal prediction objectives further compel vision-only latent states to encode unobservable physical properties by regenerating absent sensor streams from a compressed fusion bottleneck [140]. To preserve spatial coherence beyond flat vector embeddings, stochastic latent transition models decode fused representations into actionable 3D geometric structures and raw sensor data using GRU dynamics and RangeView LiDAR forecasting [15]. Finally, token-level self-introspective decoding mechanisms prune low-importance visual signals within early decoder layers via rank-correlation filtering, effectively suppressing hallucinations during generative text synthesis without altering input embeddings [77].

Architectural Efficiency and Objective Balancing. By deploying decoupled expert feedforward networks within a shared attention mechanism, unified latent architectures bypass pixel-space mediation bottlenecks to prioritize semantic alignment over reconstruction fidelity [83]. This structural design utilizes discrete semantic tokens and interleaved reasoning loops, often offloading diffusion decoding to maintain computational tractability while preserving high-level coherence. Complementing this architectural strat-

egy, hybrid objective functions dynamically assign contrastive mutual information estimators to high-dimensional noisy modalities while retaining reconstruction losses for low-dimensional proprioceptive inputs [12]. Such selective loss allocation, exemplified by Variational-CoRAL and Predictive-CoRAL frameworks integrated with Recurrent State Space Models, effectively mitigates representation collapse by balancing distraction filtering against the precision required for temporal dynamics propagation.

3.4.4 Modality-Agnostic Encoders

Unified Tokenization Schemes for Heterogeneous Input Formats. By normalizing diverse data structures into a shared 1D token sequence, architectures can employ a single frozen transformer backbone to process heterogeneous inputs without modality-specific branches [218]. Alternative strategies transpile low-density visual and tabular features directly into high-density natural language representations, effectively decoupling modality-specific perception from downstream reasoning tasks [16]. Further extending this autonomy, reference frameworks project distinct spatio-temporal token dimensions into a unified language feature space while preserving modality independence through targeted prompt engineering [137].

Pretraining on Unpaired and Asymmetric Data. Sequential training strategies group tasks by complexity and mix modalities within mini-batches to induce regularization, enabling cross-modal knowledge sharing from unpaired visual, audio, text, and 3D sources through masked pretraining [141]. To resolve scaling asymmetries between the disparate data volume requirements of visual modalities and the parameter density of language models, hybrid objectives combine autoregressive next-token prediction with flow-matching diffusion via sparse expert routing that directs modality-specific inputs to specialized sub-networks [153]. Variance-covariance regularization further stabilizes single-stream early fusion by enforcing intra-modal semantic consistency while minimizing inter-modal distances, thereby preventing feature domination and model collapse without requiring paired examples [146]. Finally, modality-agnostic permutation learning operates on discrete unified representations to enhance open-set generalization, aligning features at both holistic and fine-grained temporal levels through quantized codeword segments and sample-specific masking [71].

Architectural Mechanisms for Mitigating Modality Interference. To prevent feature contamination within a unified backbone, sparse mixture-of-experts

layers employ dynamic token-level routing that isolates modality-specific priors during bidirectional interaction [184, 215]. This decoupling strategy activates only a top- k subset of experts per token, effectively segregating distributional gaps while maintaining parameter sharing across the feed-forward network. Complementing this structural sparsity, explicit cognitive mediator tokens and task-aware instance association frameworks introduce intermediate representation buffers that separate raw sensory encoding from cross-modal reasoning [184, 215]. These injected tokens function as transient state buffers, preserving calibrated identity information and enhancing spatiotemporal modeling through memory-enhanced modules without conflating distinct input modalities. Furthermore, adaptive centroid-based anchoring replaces fixed reference modalities to symmetrically optimize intra-modal sufficiency and inter-modal alignment [79]. By generating dynamic centroids via batch-wise aggregation, this approach eliminates bias toward any single modality in the shared latent space, thereby stabilizing convergence through symmetrized mutual information maximization.

Scalability and Parameter Efficiency in Task-Specific Adaptation. Reformulating heterogeneous human-centric tasks as modality translation problems allows diverse output heads to condense into fundamental semantic classification and location regression blocks, effectively eliminating task-specific parameters while utilizing a shared encoder-decoder backbone with learnable modality indicators [168]. To further align inter-modal correlations within a single segmentation architecture, knowledge distillation from multi-modal vision-language teachers enables dynamic selection of fine-grained features via cross-modal similarity maps, ensuring robust performance regardless of input modality availability [230]. Addressing the challenge of divergent tracking modalities, serial task integration replaces parallel data mixing with replay-based continual learning mechanisms that preserve historical distributions to mitigate catastrophic forgetting while maintaining a unified parameter set across the system [149].

3.4.5 Disentangled Representation Learning

Information-Theoretic Constraints for Latent Factor Independence. Variational decomposition frameworks explicitly isolate redundancy, uniqueness, and synergy by factorizing multimodal representations into intra-modality and inter-modality components governed by information-theoretic bounds [199]. These approaches maximize mutual information to define distinct R , U , and S terms while employing dynamic gating mechanisms to model sample-specific

interactions without conflating shared semantics with modality-private noise. Moving beyond static factorization, hierarchical semantic residual mechanisms replace numerical subtraction with mutual information minimization to separate modal-general semantics from fine-grained features across encoder layers [70]. Such architectures integrate estimators like CLUB and Cross-Modal CPC within two-layer hierarchies, often utilizing discrete codebook quantization to enforce strict orthogonality between global concepts and local residuals. For domains requiring physical consistency, product-of-experts fusion integrates parametric domain models directly into mixture-of-experts decoders to structure latent clusters according to known physical states [154]. By embedding Gaussian Mixture Model priors and specific expert functions into the likelihood term, these methods optimize a closed-form evidence lower bound via Expectation-Maximization, ensuring that disentangled factors adhere to underlying dynamical constraints rather than purely statistical correlations.

Adversarial and Distributional Decoupling. Explicit suppression of domain-specific artifacts during contrastive alignment enforces modality-invariant representations by minimizing mutual information between latent features and sensor identity [236]. This adversarial decoupling utilizes gradient reversal within a Sensor-Aware Modulator to route weights that isolate tactile features from their source, ensuring the learned embedding space remains agnostic to hardware variations. Distinct from deterministic point estimates, probabilistic frameworks further separate training dynamics from inference constraints by optimizing distributional parameters rather than fixed coordinates [173]. By sampling from multivariate Gaussian distributions and applying variance regularization, these methods relax rigid intra-class directional alignment requirements while retaining mean vectors for prediction tasks. The integration of hard combination regularizers allows the model to maintain non-parallel inference embeddings, effectively decoupling the stochastic nature of the training phase from the deterministic needs of downstream applications.

Disentanglement as a Regularizer for Few-Shot Generalization. Constructing unified category-centric representation spaces allows the synchronized application of standard domain generalization augmentations by isolating general semantic factors from modality-specific noise [72]. This separation facilitates techniques such as supervised contrastive learning and mutual information minimization to enforce category-general features while suppressing irrelevant components during tasks like Mixup or Jigsaw puz-

zles. In few-shot in-context scenarios, stabilizing performance requires decoupling demonstration encoding from backbone activations to mitigate cross-modal interference [189]. Mechanisms employing low-rank transformations and adaptive dense routing modulate prototypes dynamically, preventing the non-monotonic degradation often observed with increasing shot counts. Furthermore, enforcing strict orthogonality between latent factors acts as an intrinsic structural regularizer that curbs catastrophic forgetting in class-incremental settings [21]. By maintaining disjoint subspaces for shared semantics and private variations, these approaches preserve prior semantic clusters without overwriting them, ensuring that new classes are integrated through factor separation rather than knowledge fusion.

Controllable Editing via Explicit Factor Manipulation. Architectural decomposition assigns distinct conditioning streams to ego-motion, object dynamics, and pose factors, enabling independent manipulation of scene elements without cross-factor interference [60]. By injecting sparse DINOv2 tokens alongside identity embeddings through dedicated ObjectNet and PoseNet modules, these frameworks utilize cross-attention LoRA adapters to isolate ego-trajectories from static content. Complementing this structural separation, learnable MLP projectors resolve multi-expert conflicts by enforcing representational orthogonality across heterogeneous semantic and geometric priors [185]. Regularization via Centered Kernel Alignment and cosine similarity losses ensures that frozen foundation experts maintain subspace alignment, effectively decoupling modality-specific features to prevent feature conflict during generation.

3.4.6 Discussion

Achieving true modality-agnostic efficiency frequently necessitates a compromise where aggressive latent compression or high-density textualization erodes the fine-grained perceptual fidelity essential for dense prediction tasks [16, 61, 129, 230]. While such architectural streamlining accelerates inference, it often introduces cascading errors in captioning or creates gaps in pixel-level segmentation that continuous latent rollouts and integer quantization schemes struggle to recover. Conversely, coupling disentangled representation learning with symbolic knowledge structures offers a pathway to mitigate these deficits by separating content from style and grounding abstract concepts within relational graphs [21, 39, 82, 173]. This integration facilitates zero-shot generalization through probabilistic embedding sampling and hierarchical similarity constraints, effectively anchoring

policy optimization to scene concept graphs. Despite these advances, significant barriers persist in aligning heterogeneous expert representations without inducing feature conflict or incurring prohibitive computational overhead for orthogonality enforcement [185, 222]. Current approaches relying on Centered Kernel Alignment regularization or neural gating mechanisms often face trade-offs between retrieval coverage dependency and the compute resources required to filter gating noise, leaving the goal of seamless, conflict-free multimodal fusion partially unrealized.

3.5 Discussion

A structural convergence is reshaping the field, where generative world modeling and discriminative policy learning increasingly merge into unified token-based architectures that share latent spaces for both simulation and control [37, 60, 227]. This unification relies on modality-agnostic representation strategies that decouple semantic reasoning from sensor-specific encoding, enabling robust cross-modal transfer and sparse data generalization through high-density textualization and dual embodied-symbolic fusion [16, 21, 134, 230]. Despite these architectural advances, significant scaling bottlenecks persist due to the computational cost of sequential simulation rollouts, dependency on upstream perception accuracy, and latency introduced by explicit reasoning loops [6, 37, 60, 217].

4 Challenges and Prospects

This chapter critically examines the persistent obstacles hindering the deployment of robust multimodal world knowledge systems and outlines a strategic roadmap for future research. Sec. 4.1 identifies and deconstructs the fundamental technical and theoretical barriers currently limiting the fidelity, efficiency, and reliability of these models. Following this analysis, Sec. 4.2 details concrete future research directions and emerging paradigms designed to resolve current limitations. Together, these subsections provide a comprehensive assessment of the field’s trajectory toward Artificial General Intelligence.

4.1 Challenges

The development of reliable multimodal world models is currently constrained by a convergence of fundamental technical and theoretical barriers that impede their deployment in complex, dynamic environments. Primary among these obstacles are the

frequent violations of physical laws and causal logic within generated states, as detailed in [Sec. 4.1.1](#), alongside the progressive degradation of temporal coherence over extended interaction horizons discussed in [Sec. 4.1.2](#). Further complicating model fidelity is the persistent difficulty in achieving precise semantic alignment across heterogeneous modalities, a challenge explored in [Sec. 4.1.3](#), which interacts critically with the prohibitive computational costs required for scaling such architectures ([Sec. 4.1.4](#)). Finally, the field faces significant hurdles in establishing rigorous evaluation protocols that reflect real-world complexity ([Sec. 4.1.5](#)) and in acquiring the diverse, high-quality data necessary to train robust systems ([Sec. 4.1.6](#)). Addressing these interconnected limitations requires a coordinated re-examination of model architectures, training objectives, and benchmarking standards.

4.1.1 Physical Consistency and Causal Validity

Despite advances in generative fidelity, current multimodal world models frequently fail to internalize the fundamental laws governing physical dynamics, resulting in predictions that are visually plausible but mechanistically invalid. This disconnect arises because optimization objectives often prioritize statistical likelihood over causal necessity, leading to systematic violations of conservation principles and brittle reasoning chains when faced with counterfactual scenarios. The following analysis details specific structural deficits in how these systems represent and propagate physical state changes.

- 1) **Hallucinated State Transitions.** Generative architectures often produce physically impossible outcomes, such as objects violating buoyancy constraints or undergoing erroneous phase transitions, because their training losses emphasize pixel-level reconstruction over adherence to implicit conservation laws [59, 101, 143, 152]. This aesthetic bias creates a paradox where models generate high-fidelity video sequences that nonetheless exhibit spatiotemporal inconsistencies and a profound deficit in causal reasoning.
- 2) **Architectural Inductive Bias Deficits.** Purely neural autoregressive and diffusion frameworks lack the necessary inductive biases to model continuous physical dynamics, causing prediction errors to compound rapidly without explicit coupling to parametric simulators or symbolic constraints [154, 164, 207, 233]. Consequently, these systems struggle with phenomena requiring partial differential equation grounding, such as upwind propagation, and often confuse unimodal

patterns with multimodal transition logic.

- 3) **Dynamic Mechanistic Blindness.** Vision-language models demonstrate near-random performance on tasks requiring dynamic mechanistic simulation, failing to disentangle changing attributes from static perceptual features even when they possess strong scene understanding capabilities [2, 9, 49]. These models cannot reliably predict motion trajectories or discriminate counterfactual states, indicating severe limits in representing abstract variables and grounding predicates for transitive inference.

4.1.2 Long-Horizon Temporal Coherence

Temporal degradation in multimodal agents arises not merely from information loss but from the structural misalignment between autoregressive objectives and the requirements of stable state tracking. As interaction sequences extend, the compounding effect of prediction errors decouples latent dynamics from observable reality, while memory mechanisms struggle to distinguish relevant history from accumulated noise. This fragility is exacerbated when reasoning processes amplify minor grounding errors, leading to a collapse in the consistency of the agent’s internal world model over time.

- 1) **Autoregressive Drift.** Standard sequence modeling objectives optimize for next-token likelihood rather than latent state consistency, causing cumulative deviations in long-horizon planning tasks [5, 213, 219]. This mismatch leads to a failure in inferring true underlying dynamics, particularly under partial observability where the model collapses distinct states into identical predictions. Consequently, reasoning chains degrade rapidly once the interaction length exceeds a specific threshold, rendering long-term strategies unreliable.
- 2) **Reasoning-Induced State Drift.** Extended chain-of-thought processes often act as non-linear amplifiers that detach high-level semantic reasoning from the evolving temporal state of the environment [189, 196, 212]. Such cognitive overload induces a divergence where the model generates plausible but factually inconsistent narratives that contradict earlier established facts. The resulting internal interference prevents the maintenance of a coherent timeline, especially when scaling to complex, multi-step interaction sequences.
- 3) **Memory Saturation.** Many current architectures rely on append-only update strategies that

indiscriminately accumulate obsolete states, creating severe retrieval bottlenecks in dynamic environments [105, 142]. Ambiguity in defining event boundaries forces the system to retain irrelevant semantic data, leading to context queue saturation. This decoupling between retrieval utility and stored content means that critical historical information becomes inaccessible amidst the noise of past interactions.

- 4) **Sensory Evidence Overwrite.** Visual attention signals tend to decay exponentially across multi-turn dialogues, allowing textual hallucinations to progressively overwrite original sensory evidence [159, 205]. This attention reallocation failure triggers a snowballing effect where downstream agents operate on fabricated premises rather than grounded observations. The suppression of visual tokens in favor of dominant textual patterns erodes mid-layer features essential for maintaining factual accuracy.

4.1.3 Cross-Modal Semantic Alignment

Achieving semantic unity across text, vision, and action spaces encounters fundamental barriers where optimization dynamics and representational mismatches actively degrade alignment quality. Even when architectures theoretically support multimodal fusion, the learning process often collapses into suboptimal states where one modality dominates or logical traces drift from sensory evidence.

- 1) **Optimization-Induced Modality Laziness.** Joint training objectives frequently cause dominant modalities to suppress unique feature learning in weaker streams, leading to quantity laziness and interaction deficits [43, 199, 226]. This redundancy suppression creates static fusion noise and reinforces teacher exclusion, preventing the model from forming a truly variational decomposition of cross-modal information.
- 2) **Desynchronization in Explicit Reasoning Chains.** Attempts to bridge perception and logic via explicit reasoning often trigger temporal drift, causing semantic traces to lose synchronization with raw sensory inputs [178, 212, 239]. Such visual over-interpretation propagates errors through rigid linear chains, resulting in inertial numeric guessing and attention dilution that overwhelms cognitive capacity.
- 3) **Structural Mismatch Between Discrete and Continuous Spaces.** Persistent alignment gaps arise from the inherent conflict between discrete symbolic abstractions and continuous pixel

spaces, which resist standard tokenization or late-fusion strategies [57, 117, 134, 230]. These structural frictions manifest as allocentric-egocentric biases, topology abstraction failures, and latent trajectory mismatches that hierarchical routing bottlenecks cannot resolve.

- 4) **Fragility of Contextual Grounding Mechanisms.** Grounding systems remain highly susceptible to misleading cues or decorative visuals, suffering significant performance drops when semantic priors contradict evidential data [65, 80, 113, 191]. This vulnerability leads to traceability errors and entity extraction dependencies, particularly when facing misleading clinical contexts or cultural value disparities.

4.1.4 Computational Efficiency and Scalability

The deployment of large-scale multimodal world models faces severe constraints due to the compounding costs of high-resolution spatial modeling and dense temporal processing. Current architectures often incur prohibitive latency during inference while consuming excessive energy during training, primarily because they lack mechanisms to adaptively sparsify computation based on scene dynamics or task complexity. These inefficiencies stem from fundamental design choices that prioritize representational capacity over operational feasibility, creating barriers to real-time application and sustainable scaling.

- 1) **Sequential Visual Decoding Latency.** Autoregressive generation of visual tokens imposes a strict sequential dependency that scales linearly with spatial resolution, rendering real-time simulation impractical for high-fidelity environments. The necessity to decode thousands of tokens per image frame, such as the 1024 tokens required by certain tokenizers, introduces cumulative delays that prevent interactive response rates [33].
- 2) **Uniform Computational Expenditure.** Static architectural designs enforce dense module activation across all timesteps, failing to reduce processing load during periods of trivial motion or redundant state transitions. This inability to conditionally route computation results in quadratic attention complexity even when scene dynamics are minimal, wasting resources on uninformative frames [114, 148].
- 3) **Verbalized Reasoning Overhead.** Explicit chain-of-thought mechanisms consume disproportionate context window capacity and decoding cycles compared to compact latent representations. Generating extensive textual rationales

for every decision step creates a bottleneck in both memory usage and inference speed, limiting the horizon of feasible rollouts [61, 200].

4) **Unsustainable Pre-training Demands.**

Achieving high-fidelity world modeling requires training corpora exceeding trillions of tokens, which renders standard Transformer scaling laws economically and energetically unsustainable without significant architectural compression. The reliance on massive data-driven autoregressive scaling exacerbates energy consumption, highlighting a critical gap between model performance and ecological viability [33, 114].

4.1.5 Evaluation and Benchmarking Fidelity

Current evaluation frameworks for multimodal world models frequently yield inflated performance estimates by relying on static, curated datasets that fail to replicate the stochastic noise and distributional shifts of physical deployment. This disconnect is compounded by metric designs that prioritize task completion over genuine state comprehension, allowing agents to exploit superficial heuristics while masking fundamental reasoning deficits. Furthermore, standard protocols often conflate instruction compliance with logical competence, obscuring the suppression of critical flaw detection mechanisms under alignment pressures.

- 1) **Synthetic Data Artifacts.** Benchmarks constructed from digitally rendered or heavily curated inputs systematically overestimate model capabilities because they lack the distributional shift inherent in real-world capture conditions. Studies indicate that accuracy degrades significantly when models encounter print artifacts, screen distortions, or lighting variations absent from training distributions, revealing a fragile reliance on clean data rather than robust environmental modeling [80, 87, 89, 98].
- 2) **Proxy Metric Insufficiency.** Conventional success metrics act as poor proxies for environment understanding, often masking an agent’s reliance on heuristics while failing to detect deficits in latent property acquisition. Trajectory matching and terminal-state verification frequently diverge from true comprehension scores, particularly under partial observability constraints where agents can achieve high rewards without maintaining coherent internal state representations [48, 93, 107].
- 3) **Compliance-Reasoning Conflation.** Evaluation protocols often mistake adherence to in-

structions for reasoning competence, obscuring the suppression of implicit flaw detection mechanisms due to alignment pressures favoring affirmative responses. When faced with instruction-scene misalignments or cultural value conflicts, models tend to prioritize compliance over correcting errors, indicating a fragility in handling implicit conflicts that standard benchmarks fail to probe [121, 191, 192].

- 4) **Compositional Granularity Gaps.** Existing assessment frameworks lack the granularity to identify semantic coverage gaps, preventing the precise localization of long-tail concept deficiencies within multimodal training distributions. Without atomic proposition extraction or difficulty-guided sampling, evaluations miss compositional retrieval failures and cannot adequately measure performance in open-world tool integration scenarios [22, 62].

4.1.6 Data Scarcity and Quality

The viability of multimodal world models is fundamentally constrained by a compounding crisis where data scarcity, annotation noise, and domain fragility intersect to limit generalization capabilities. While passive video corpora offer vast quantities of visual tokens, they lack the actionable interaction trajectories required for embodied agents to learn causal dynamics effectively [27, 93, 100]. This asymmetry forces reliance on offline pretraining that fails to capture sparse reward signals, creating a persistent domain gap when transferring policies from simulation to real-world environments.

- 1) **Actionable Data Asymmetry.** The abundance of passive observation data contrasts sharply with the scarcity of annotated interaction logs, creating a hard ceiling on the ability of models to generalize beyond static prediction. Existing datasets rarely provide the dense, step-wise feedback necessary for learning complex manipulation tasks, leaving agents ill-equipped to handle the stochasticity of physical engagement [100]. Consequently, models trained on such imbalanced corpora often hallucinate plausible-looking but physically invalid action sequences when deployed in dynamic settings.
- 2) **Synthetic Data Fragility.** Reliance on procedurally generated environments introduces subtle physical inaccuracies that propagate as systematic biases during policy learning [80, 87]. Although synthetic augmentation expands dataset volume, the simplified dynamics fail to capture the high-frequency contact variations present

in real-world robotics, leading to catastrophic failure upon deployment [17, 130]. These discrepancies force models to overfit to simulator-specific artifacts rather than developing robust latent representations capable of adapting to novel physical constraints.

3) Annotation Sparsity and Weak Supervision.

The absence of dense, step-wise action annotations in current video datasets compels models to depend on noisy, weakly-supervised signals that degrade causal learning fidelity. Without explicit trajectory labels, training objectives often conflate correlation with causation, preventing agents from distinguishing between incidental background motion and critical state changes driven by intervention. This reliance on implicit cues results in brittle policies that falter when faced with the precise temporal granularity required for fine-grained control tasks.

- 4) Scaling-Induced Cognitive Bias.** Indiscriminate expansion of training corpora introduces significant redundancy and reinforces language-prior biases that obscure rare but critical cognitive samples [91, 99]. Filtering mechanisms designed to mitigate this noise frequently exclude the most challenging examples required for robust reasoning, such as those with zero-pass rates in initial validation [18]. This selective pruning creates a feedback loop where models become increasingly proficient at common patterns while remaining incapable of resolving complex, out-of-distribution anomalies.

4.2 Prospects

The path toward Artificial General Intelligence requires addressing current architectural and operational constraints through several converging research trajectories. [Sec. 4.2.1](#) examines the fusion of neural learning with symbolic reasoning to improve logical consistency, while [Sec. 4.2.2](#) proposes autonomous systems capable of curating their own training data. The field anticipates a shift toward [Sec. 4.2.3](#), which details architectures designed for native multi-modal processing without specialized adapters. Concurrently, [Sec. 4.2.4](#) and [Sec. 4.2.5](#) focus on aligning world models with human values and integrating them into physical robotic embodiments. Finally, [Sec. 4.2.6](#) addresses the critical need for sustainable computing practices to support the energy demands of these advancing systems.

4.2.1 Neuro-Symbolic Integration

Architectural decoupling offers a direct path to verifiable causal reasoning by separating world modeling from language-based inference. Future systems must isolate causal dynamics within dedicated modules to prevent large language models from hallucinating during multi-step planning tasks [51, 228]. This separation relies on disentangled latent interfaces and text-based action encodings that enable symbolic sub-goal validation across modular perception-planning-execution pipelines. Explicit state representations employing immutable logical structures provide the architectural guarantees necessary for constructing composable counterfactuals, a capability implicit neural weights fundamentally lack. Such designs utilize fixed structural definitions with deterministic logging mechanisms to ensure stable rule evaluation and consistent definition chains. Hybrid inference mechanisms that dynamically switch between symbolic path planning and neural policies further enhance generalization in few-shot embodied scenarios [51, 228]. These frameworks integrate efficient search operations with autoregressive latent trajectory prediction, allowing tree search operations over simulators guided by commonsense logic predicates.

Replacing dense vector embeddings with explicit propositional codebooks will enable compositional retrieval and precise identification of semantic coverage gaps in future multimodal systems. Such architectures rely on atomic proposition extraction and structured grouping to construct canonical codebooks that support logical query operations for analyzing long-tail concept deficiencies. Reliable human-robot collaboration further demands dynamic common ground models that integrate procedural knowledge graphs with neural modules to ensure verifiable causal updates. These frameworks utilize neuro-symbolic continual learning loops where inductive large language model inference couples with deductive symbolic validation to represent shared intentionality and sensorimotor experiences. Dual-level representations fusing embodied sensorimotor features with amodal concept graph embeddings mitigate catastrophic forgetting while enhancing cross-modal semantic alignment through mechanisms that unify graph structures with neural latent spaces [21]. Probabilistic reasoning frameworks grounded in unified symbolic state-action sequences significantly improve theory of mind reasoning under partial observability compared to black-box approaches by aligning video scene graphs with text-derived predicate parsing.

Counterfactual simulation under uncertainty will likely evolve through hybrid inference frameworks

that couple neural hypothesis generation with symbolic validation routines to enforce physical constraints. Such architectures parse multimodal cues to satisfy stochastic geometric rules, effectively refining probability distributions within uncertainty-aware hypothesis spaces while managing non-linear sensory correlations. To optimize reasoning efficiency, agents may selectively invoke external world modules as inspectable visual artifacts, enabling the validation of temporal dynamics through sparse tool calling and visual thought experiments [122]. Reinforcement learning can then balance reward signals against informational gain to determine dynamic rollout fidelity during these processes. Furthermore, scalable planning emerges when large models integrate behavioral priors derived from smaller, specialized controllers, facilitating multi-step execution without full parameter fine-tuning. This integration aligns internal state representations to produce calibrated inference-time outputs, ensuring accurate stepwise transition modeling across varying scales of knowledge transfer.

4.2.2 Self-Evolving Data Ecosystems

Transitioning from static curation to autonomous synthesis, future data ecosystems will enable agents to construct their own training curricula through interactive environmental exploration. By employing self-instruct mechanisms and scaling synthetic task complexity, these systems can accumulate experience without immediate gradient updates while utilizing information obfuscation strategies to deepen representational learning [90, 171]. Dynamic sampling protocols will further refine this process by prioritizing high-uncertainty scenarios, effectively bootstrapping reasoning capabilities on sparse-reward tasks through structural ambiguity injection. Continuous performance gains will subsequently rely on retrieving and refining multimodal memories of successful trajectories rather than depending solely on parameter adjustments. This approach integrates situation-aware planning loops with backward task decomposition to index visual-textual experiences for efficient reuse [171].

To prevent the accumulation of prediction errors during autonomous data synthesis, future ecosystems will integrate internal evaluators that verify physical consistency within generative feedback loops [27]. These agent-evaluator-model cycles enforce causal validity and correct manifold drift by filtering low-quality synthetic outputs to maximize sample efficiency. Complementing this verification, cognitive reflection mechanisms will extract task-relevant dynamics from synthetic trajectories to ensure semantic alignment with environmental constraints [50]. Such systems utilize

cross-attention grounding and working memory integration to filter synthetic instructions before they enter the training corpus. Progressive curriculum strategies will further stabilize learning on these self-generated multimodal sequences by evolving from simple unimodal transitions to complex cross-modal interactions [50]. This approach mitigates loss escalation through dream tuning paradigms that scale modality combinations gradually. The convergence of iterative quality assurance protocols and cognitive augmentation will ultimately establish closed-loop frameworks where agents refine their own physical priors [27, 50].

Preventing distributional collapse in autonomous data generation requires explicit divergence constraints to halt retreat into narrow, self-reinforcing loops that lose environmental grounding [27, 90]. Without such safeguards, the combined effects of manifold drift and informational ambiguity saturation threaten to degrade synthetic curricula over time. Maintaining causal validity during self-evolution further demands a dynamic equilibrium between generated complexity and verifiable physical priors [27]. Architectures must therefore embed feedback mechanisms that enforce consistency by filtering outputs through inverted pyramid workflows. Task-aligned robotic data validation serves as a critical mechanism to ensure that synthetic expansions remain tethered to observable reality rather than drifting into hallucinated regimes.

4.2.3 Native Omni-Modal Architectures

Abandoning modality-specific adapters, next-generation frameworks will map diverse sensory inputs and actions directly into a single geometric latent space through unified tokenization schemes [1, 137]. This structural convergence enables early fusion of semantic and visual signals within a shared backbone by employing cascaded encoding structures that support joint autoregressive and diffusion training [1, 110]. Such architectures utilize layer-wise early fusion and bidirectional context attention to integrate understanding and generation pathways without requiring separate processing streams. To eliminate inherent visual bias, these systems will align all modalities to LLM-augmented semantic centers rather than relying on image-centric anchors [111, 137]. The resulting representation spaces achieve balance by localizing embedding centers around rich textual semantics, ensuring that language serves as the primary reference framework while preserving modality autonomy.

Addressing scaling asymmetries between data-hungry

vision tasks and parameter-intensive language objectives requires Mixture-of-Experts architectures that employ modality-specific routing to allocate distinct parameter subsets dynamically [35, 153]. Such sparse gating mechanisms facilitate expert specialization for generation versus understanding, effectively harmonizing divergent scaling laws within a single backbone. Complementing this structural sparsity, hybrid training protocols utilize dual-tower structures to decouple autoregressive reasoning from diffusion-based generation, preventing gradient interference while maintaining shared semantic representations [1, 35]. These designs separate causal and bidirectional contexts through dual-stream joint attention and rectified flow integration to mitigate objective conflicts. Stabilization in these unified systems further depends on regime-specific optimization strategies, including differential learning rates and modality-mixture scheduling, which manage competing loss landscapes during convergence [8, 153]. Techniques such as progressive unfreezing schedules and QK-Norm stabilization prove essential for maintaining training stability across diverse modalities. Finally, on-policy distillation mechanisms consolidate multi-teacher capabilities into single native models, enabling symmetric any-to-any modeling without architectural rewiring [1, 8].

Theoretical frameworks posit that scaling multitask training compels distinct architectures to converge toward a single, modality-agnostic statistical model of reality rather than maintaining separate representational manifolds [76]. This alignment stems from simplicity biases in large-scale models, where Pointwise Mutual Information kernel approximation drives the reduction of representational variance and enforces cross-modal linearity. Future benchmarking protocols must therefore enforce strict cross-modal complementarity through ablation-based screening to validate genuine fusion capabilities [81]. Such evaluations require bidirectional full-modality input-output configurations that eliminate unimodal cues, ensuring metrics reflect fusion-dependent reasoning rather than semantic fidelity derived from single sources.

4.2.4 Human-Centric Alignment and Safety

Adopting hierarchical training curricula that decompose cognitive development into factual, conceptual, and procedural stages offers a viable path toward value-grounded reasoning in future world models [54]. This structured layering aligns with established educational taxonomies to enforce sequential knowledge acquisition, thereby reducing hallucinations through rigid progression constraints. By integrating explicit knowledge-driven chain-of-thought structures, agents

can map specific cognitive steps to verified information domains during inference [54]. Such segmentation enables procedural execution grounding where intent verification protocols actively filter unsafe decisions before action selection. The resulting framework ensures that multimodal input fusion occurs only within boundaries defined by pre-validated safety constraints.

Moving beyond binary safety classifiers, future frameworks must establish quantitative metrics to evaluate the degree of social norm adherence within generated multimodal trajectories. This shift requires developing scoring mechanisms that measure norm deviation, analyze trajectory probability distributions, detect implicit biases, and apply contextual appropriateness weighting to assess behavioral nuances. Integrating human feedback loops directly into the latent dynamics of world models will further enable continuous calibration of predicted behaviors against evolving societal expectations. Such integration facilitates preference-based reward shaping, latent space regularization, dynamic constraint injection, and real-time alignment correction to maintain fidelity with human values.

4.2.5 Embodied General Intelligence

Achieving human-like physical agency requires migrating from reactive policy mappings to explicit, rollout-capable world simulators that support counterfactual reasoning. This structural shift decouples latent dynamics from control objectives, enabling agents to perform anticipatory planning through internal simulation rather than relying on immediate, reflexive responses [123]. Integrating visual generation directly into these reasoning chains resolves the spatial fidelity bottlenecks inherent in verbal-only models, providing the geometric precision necessary for complex manipulation tasks [179]. Such interleaved verbal-visual architectures allow systems to simulate atomic physical interactions with high accuracy before executing actions in the real environment. Advanced training protocols must further compress multi-turn interactive exploration into compact, single-turn internal representations to foster proactive behavior [139]. These frameworks transform standard optimization procedures into a mechanism for learning dense, predictive world models capable of long-horizon foresight without relying on specific loss function manipulations.

Converging physical control with generative world modeling requires unifying continuous motor commands and discrete linguistic tokens within a single representational space. Recent proposals suggest

translating actions into abstract symbolic units to enable joint training alongside visual generation objectives using modality-agnostic transformer backbones [195, 227]. To maintain consistency between predicted environmental dynamics and executed policies, future architectures will likely employ a unified decoding mechanism that simultaneously produces future state trajectories and optimal control signals in one forward pass [187, 227]. This structural evolution supports parallel processing streams where specialized components handle hierarchical state representations while continuous diffusion processes condition latent dynamics. Bridging the gap between passive video understanding and active manipulation further demands surrogate temporal tasks that predict visual trajectories interleaved with explicit action tokens [187, 195]. Such frameworks utilize spatial grounding techniques to align action magnitudes with visual scales, enabling offline reinforcement learning on simulated rollouts without decoupled planning modules.

Embedding explicit interoceptive variables, such as energy levels and thermal stress, as latent states within predictive loops offers a pathway to human-like generalization by dynamically modulating reward landscapes and action planning [84]. This homeostatic grounding transforms agents from passive observers into coupled dynamical systems where internal state regularization imposes existential constraints on behavior. Current evaluation protocols must consequently expand beyond external manipulation metrics to include benchmarks for internal state regulation and prosocial alignment derived from shared homeostatic sensitivity [84]. Addressing existing gaps in affordance recognition and spatial reasoning requires simulated environments that test these interoceptive awareness metrics rather than relying solely on narrow perceptual proxies that fail to capture physiological self-monitoring.

4.2.6 Sustainable and Efficient Computing

Conditional computation strategies offer a direct path to reducing the carbon footprint of training future world models by dynamically activating heavy generative modules only during critical physical transitions. This approach bypasses expensive video forecasting for routine actions through token-level dynamic routing and sparse inference patterns [148]. Further efficiency gains may arise from replacing direct high-dimensional visual input with structured textual encodings of environmental states, potentially lowering information overload and accelerating planning convergence in embodied agents [56]. Integrating explicit future state prediction into the reasoning loop enables agents to prune inefficient action trajectories

early, minimizing redundant computational cycles during long-horizon tasks via lookahead simulation mechanisms [56, 148].

Transitioning from von Neumann bottlenecks requires adopting event-driven neuromorphic substrates that activate only upon detecting physical state transitions. Such architectures utilize spike-based encoding and asynchronous event streams to eliminate static power consumption during idle periods, ensuring energy is expended solely when temporal contrast sensing indicates a change in the environment. Extending this efficiency to latent dynamics models involves deploying spiking neural network variants directly onto analog memristive crossbars. This approach facilitates in-memory computing where conductance-based synaptic weights perform matrix-vector multiplication without moving data between separate storage and processing units. Continuous low-power inference becomes feasible for embodied agents operating in resource-constrained settings through this co-location of computation and memory.

Hardware-aware algorithmic sparsity provides a direct mechanism to reduce carbon emissions by dynamically pruning network activations when input redundancy exceeds a defined threshold, enabling energy-proportional inference [144]. Complementing this dynamic routing, decomposing complex evaluations into atomic verification steps isolates computational bottlenecks early in the pipeline, thereby preventing redundant full-model invocations during routine processing [59]. Training lightweight adapters on negative contextual data further supports this efficiency by allowing systems to detect ambiguity without invoking full generative simulation, ensuring that power consumption scales strictly with task complexity rather than total model capacity [59, 144]. These dynamic compute allocation policies facilitate computational offloading, directing high-energy resources only toward instances requiring deep analysis while maintaining low-power modes for standard operations. The convergence of these strategies points toward architectures where system resource allocation adjusts automatically based on intrinsic data characteristics.

5 Conclusion

Multimodal World Knowledge advances from static perception to active environmental modeling by converging generative simulation, embodied control, and cognitive reasoning. This transition establishes unified computational frameworks where agents learn latent dynamics and execute vision-language-action

policies rather than merely classifying inputs. Despite these architectural advances, significant barriers remain regarding physical consistency, long-horizon temporal coherence, and the computational efficiency necessary for real-world deployment. Current systems frequently struggle with hallucination mitigation, causal validity, and state tracking degradation over extended sequences, while inference constraints limit scalability. Future progress demands a strategic integration of neuro-symbolic reasoning, self-evolving data ecosystems, and native omni-modal systems to achieve reliable Artificial General Intelligence. Achieving this objective requires fusing symbolic logic with neural learning, implementing autonomous data curation pipelines, and developing modality-agnostic encoders governed by strict human alignment protocols. The community must establish standardized benchmarks and open-source frameworks to bridge the gap between theoretical designs and practical application.

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